Uniqueness of special Lagrangian tangent cones with immersed special Legendrian links

Yang Li

arxiv:2610.10790; CC BY 4.0

Allowed standard prerequisites (editorial).

- Classical geometric research prerequisites cited by the author. Complete original embedded bibliography retained. Standard prerequisites remain cited; all source-local carrying estimates, constructions and non-adjacent proof continuations are retained.

1 Introduction

A fundamental problem in minimal surface theory is to prove uniqueness of the tangent cone at a singular point of an area-minimizing current. Three basic approaches are as follows.

1. *Allard–Almgren [2].* Suppose that one tangent cone has multiplicity one and a smooth compact link, and assume as an additional hypothesis that every Jacobi field of the link is integrable through minimal submanifolds of the sphere. On fixed annuli, regularity gives strong convergence of the normalized normal graphs to a Jacobi field. Integrability provides a nearby minimal cone corresponding to its degree-one part. Relative to this comparison cone, estimates for the remaining components on successive annuli give uniqueness of the tangent cone and convergence at a polynomial rate.

2. *Simon's Łojasiewicz method [13].* For a multiplicity-one tangent cone with a smooth compact link, analyticity of the area functional and an infinite-dimensional gradient inequality control the accumulated change under rescaling. This gives uniqueness without the Allard–Almgren integrability hypothesis, though the general rate need not be polynomial.

3. *Simon's cylindrical-cone method [14, 15].* For a multiplicity-one cylindrical tangent cone $C \times \mathbb{R}^k$ in codimension one, the link is singular along the spine. Simon proves uniqueness and polynomial convergence when C is smooth away from the vertex, strictly stable and strictly minimizing, and satisfies his conditions on homogeneous Jacobi fields: the fields of degrees zero and one are generated by translations and rotations, and certain negative integral homogeneities are absent. Nonconcentration near the spine then makes Jacobi-field approximation and a three-annulus argument effective.

Székelyhidi proves uniqueness for the nonintegrable cylinder $C(S^3 \times S^3) \times \mathbb{R}$ [16]. In place of nearby minimal cones, he constructs minimal comparison hypersurfaces on large annuli as logarithmic perturbations of cones. His nonconcentration estimate, measured in an adapted L^∞ distance, bounds the error relative to a comparison surface on an interior annulus by the graphical error away from the singular axis, plus an arbitrarily small multiple of the error

on the larger annulus. Related estimates enter his strong unique continuation theorem at cylindrical tangent cones [17]. Collins–Li adapt the strategy to exact special Lagrangians: if a multiplicity-one tangent cone is $\mathbb{R}^k \times \mathbf{C}$, where $\dim \mathbf{C} > 2$, the link of $\mathbf{C}$ is smooth and connected, and $\mathbf{C}$ satisfies their integrability hypothesis, then the tangent cone is unique [3]. Their argument uses local potentials together with nonconcentration and decay estimates.

We consider a closed integral three-current L of multiplicity one in $\mathbb{C}^3$, which is *special Lagrangian* with respect to the standard Euclidean structure (J, ω, Ω), namely

$$\omega|_L = 0, \qquad \operatorname{Im} \Omega|_L = 0.$$

By an observation of Harvey and Lawson [7, Theorem II.4.2], the fact that L is calibrated by $\operatorname{Re} \Omega$ implies that L is locally volume minimizing. We study special Lagrangians whose tangent cones have immersed surface links with transverse double points. The simplest example is the union of two 3-planes meeting along a line.

Our main contribution is a new mechanism which proves the uniqueness of the tangent cone even when L^2 concentration occurs. The normalized height may concentrate near the rays through the double points of the link, and hence disappear from the Jacobi field limit on regular regions. Instead, we turn a gluing obstruction into a discrete Łojasiewicz inequality. This controls the drift of the best approximating cone over scales.

We identify an immersed Legendrian with its image unless its domain is needed. Let $\Lambda \subset S^5$ be a compact oriented smooth immersed special Legendrian surface of phase zero, and let $F : \Sigma \to \Lambda$ denote its parametrization. The domain Σ may be disconnected even when its image Λ is connected. We allow finitely many double points, transverse in the contact distribution, and no higher intersections. Write $\mathbf{C}_0 = \mathbf{C}(\Lambda) = \{Rp : R \geq 0,\ p \in \Lambda\}$ for its special Lagrangian cone.

Write $\Sigma = \coprod_{v \in V} \Sigma_v$. The *intersection graph* $G(\Lambda) = (V, E)$ has one vertex for each connected component Σ_v and one unoriented edge for each double point, joining the vertices corresponding to the components containing its two preimages; loops and parallel edges are allowed.

Definition 1.1. We say that Λ satisfies the *intersection condition* if every edge of $G(\Lambda)$ belongs to at most one simple cycle. (Equivalently, each connected component of $G(\Lambda)$ is a *cactus graph*. A *simple cycle* is a closed path with no repeated edges and no repeated vertices except its starting and ending vertex. Loops and pairs of parallel edges are regarded as simple cycles of length one and two, respectively.)

We assume that Λ is *Jacobi-integrable*: there are a small finite-dimensional parameter neighbourhood $\mathcal{M}$ of 0 and a family $\{\Lambda_s\}_{s \in \mathcal{M}}$ of special Legendrian immersions, with $\Lambda_0 = \Lambda$, whose infinitesimal normal variations span the smooth Jacobi fields on Σ. Choose smooth parametrizations $F_s : \Sigma \to \Lambda_s$, and write $\mathbf{C}_s = \mathbf{C}(\Lambda_s)$. The family need not preserve the double points; no matching condition is imposed on their two preimages.

Our main result is

Theorem 1.2. *Under these assumptions, suppose L has $\mathbf{C}_0$ as a tangent cone at the origin and Λ satisfies the intersection condition in Definition 1.1. Then $\mathbf{C}_0$ is the unique tangent cone of L at the origin.*

Example 1.3. The hypotheses hold for a union of great special Legendrian 2-spheres meeting only in transverse double points, provided $G(\Lambda)$ is obtained from a forest by replacing each edge with two parallel edges. These two edges correspond to the two antipodal double points of an intersecting pair of 2-spheres. They form the only simple cycle containing either edge, so the intersection condition holds. The smooth Jacobi fields of each sphere integrate to independent plane deformations.

A byproduct of the proof is local rigidity of the special Legendrian link.

Theorem 1.4. *Suppose Λ is Jacobi-integrable and satisfies the intersection condition in Definition 1.1. Then every special Legendrian integral two-cycle Λ' in S^5 sufficiently close to Λ in the varifold topology equals Λ_s for some $s \in \mathcal{M}$ near 0.*

1.1 The main estimates

The quantity $D(R)$ measures the normalized L^2 height from the best approximating cone:

$$D(R)^2 = \inf_{s\in\mathcal{M}} R^{-5} \int_{L\cap(B_{4R}\setminus\overline{B}_{R/4})} \operatorname{dist}(x, \mathbf{C}_s)^2 \, d\|L\|(x). \tag{1.1}$$

Put $\Theta_0 = \|\mathbf{C}_0\|(B_1)$ and

$$m(R) = R^{-3}\|L\|(B_R) - \Theta_0, \qquad E(R) = m(4R) - m(R/4). \tag{1.2}$$

When $\mathbf{C}_0$ is a tangent cone, $m \geq 0$ and $m(R) \to 0$ as $R \downarrow 0$. By the monotonicity formula (2.1), $E(R)$ is a weighted integral of the squared normal component of the radial vector. It measures the failure of radial invariance.

The principal estimates are

$$m(R) \leq C\{D(R)^2 + E(R)\}, \qquad E(R) \geq C^{-1}D(R)^{2q} \quad \text{for some } 1 < q < 2. \tag{1.3}$$

The second estimate applies also to nearby conical currents and directly implies Theorem 1.4. Together the estimates give $E(R) \geq C^{-1}m(R)^q$. Since $q < 2$, a numerical iteration makes the sum of $\sqrt{E(R)}$ over dyadic scales finite. This controls the accumulated drift of the best approximating cone and gives uniqueness of the tangent cone; see Section 5.

1.2 Proof sketch for the discrete Łojasiewicz estimate

We explain the lower bound for E in (1.3), first when $\mathbf{C}_0$ is the union of two planes meeting along a line. The comparison link Λ_s is then the union of two great special Legendrian 2-spheres. Work at unit scale and suppose that $0 < E \ll D^{2q} \ll 1$. A normalized Jacobi field limit, together with integrability, rules out the case in which the height does not concentrate near the intersection line.

In the main case of concentration, away from the intersection line L is graphical over a nearby plane pair, while near the line it has local pieces modelled on a Lawlor neck times a line. A suitable radial slice consists of graphs over the comparison spheres joined by small Lawlor necks near the two antipodal double points of the original link. Small E requires this slice to be nearly special Legendrian. These pieces of L therefore give an approximate solution to a special Legendrian gluing problem.

Let ρ be a radius comparable to the largest neck radius. The largest logarithmic coefficient of the Lagrangian potentials is then comparable to ρ^2. One can show that $\rho^2 \geq C(\varepsilon)^{-1}D^{1+\varepsilon}$ for arbitrarily small $\varepsilon > 0$ (cf. Proposition 3.5).

The gluing obstruction. On a special Legendrian surface every period of $\lambda = \iota_x\omega$ is zero. On L, $|\lambda| = |x^\perp|$, so E controls the period around a loop γ crossing one neck and returning through the other, following the graphical sheets in between. Flux balance gives opposite logarithmic coefficients at the two punctures of each sphere, to leading order. Since the loop crosses the necks in opposite directions, their contributions add, with total magnitude comparable to $\rho^2 \log(1/\rho)$.

The two comparison spheres need not intersect after deformation. Near each original double point, they are joined by a short Reeb chord. Let $g(s)$ be the signed sum of the integrals of λ along these chords. The period formula expresses $\int_\gamma \lambda$ as the sum of $g(s)$, the logarithmic neck contribution and an error. Thus

$$|g(s)| \geq C^{-1}\rho^2 \log(1/\rho) - \left|\int_\gamma \lambda\right| - o(\rho^2 \log(1/\rho)).$$

For $E \leq \rho^5$, we have $|\int_\gamma \lambda| = O(\rho^2)$, so for small ρ,

$$|g(s)| \geq C^{-1}\rho^2 \log(1/\rho).$$

The two chords are antipodal images. Since the antipodal map preserves λ, their actions agree, but the loop traverses them in opposite directions. Thus $g(s) = 0$, a contradiction.

We now discuss an alternative approach that can be adapted to the general case. The function g is analytic with $g(0) = 0$. We rule out cancellation by comparing $g(s)$ with dg/ds. Under a deformation represented by a smooth Jacobi function h, the derivative of a chord action is twice the difference between the values of h at its endpoints. Testing the flux balance against h therefore controls dg/ds. For a single simple cycle, this gives

$$\left|\frac{dg}{ds}\right| \leq C\{\rho^2 \log(1/\rho) + \rho^{-2}\sqrt{E}\}.$$

The analytic gradient inequality gives $|g|^\theta \leq C|dg/ds|$ for some $0 < \theta < 1$. The lower bound for $|g(s)|$ and the upper bound for $|dg/ds|$ contradict this when E is too small. When there are several simple cycles, the intersection condition prevents cancellation of the logarithmic terms in a suitable weighted sum of their actions. Applying the gradient inequality to this sum gives the quantitative obstruction $E \geq \rho^{q_0}$ for some $q_0 < 8$; see Proposition 4.4. The lower bound for ρ then gives $E \geq C^{-1}D^{2q}$ with $q < 2$, as required.

Acknowledgements. The author is supported by the Royal Society URF. During this work, AI suggested the graph theoretic condition for producing a gluing obstruction, which substantially simplified an earlier draft involving a linear system of equations coming from the loop period integrals and flux balancing conditions.

2 Background

2.1 The monotonicity formula and the boundary formula

Notation. For $z \in \mathbb{C}^3$ and $\rho > 0$, let $\lambda_{z,\rho}(x) = (x - z)/\rho$ and $T_{z,\rho} = (\lambda_{z,\rho})_\# T$.

Write $\|T\|$ for the mass measure of T and $x^\perp$ for the component of the position vector x orthogonal to the approximate tangent plane of T at x.

Proposition 2.1. *For a calibrated integral three-current T without boundary in B_{R_0} and $0 < R_1 < R_2 < R_0$,*

$$R_2^{-3}\|T\|(B_{R_2}) - R_1^{-3}\|T\|(B_{R_1}) = \int_{R_1<|x|<R_2} |x^\perp|^2 |x|^{-5}\, d\|T\|. \tag{2.1}$$

For almost every $R < R_0$,

$$3\|T\|(B_R) = \int_{\langle T,|x|,R\rangle} \iota_x \operatorname{Re}\Omega. \tag{2.2}$$

Here $\langle T, |x|, R\rangle$ is the current slice, with its boundary orientation.

Proof. The first identity is the monotonicity formula [1]. The second follows from calibration, the Stokes formula and $d(\iota_x \operatorname{Re}\Omega) = 3\operatorname{Re}\Omega$. □

2.2 Contact geometry and Jacobi fields

Write $\lambda = \iota_x\omega$ for the Liouville one-form. The restriction of λ to S^5 is a contact form with Reeb vector field $x \mapsto Jx$. An immersion $F : \Sigma \to S^5$ is *Legendrian* if $F^*\lambda = 0$, and *special Legendrian* if its cone is special Lagrangian. A special Legendrian immersion is minimal in the sphere [8, Sections 2 and 3.4].

We use the Laplacian $\Delta_\Sigma = \operatorname{div}_\Sigma \nabla_\Sigma$ and write $R = |x|$. On the cone, the Jacobi equation for a normal field $V = J\alpha^\sharp$ is $d\alpha = d^*\alpha = 0$ [8, Section 5.1]. A degree-one Jacobi field has the form $V = J\nabla(R^2\phi)$.

Lemma 2.2. *Finite-energy degree-two Jacobi potentials are precisely $R^2\phi$ with $\phi \in \ker(\Delta_\Sigma + 6)$. They extend smoothly across the two distinct preimages in Σ of each double point of Λ, separately on each local sheet.*

Proof. Finite energy gives $\phi \in H^1$ near each double-point preimage. These points have zero capacity in the smooth surface Σ, so the weak equation extends across each point. Elliptic regularity gives the assertions. □

By the Jacobi-integrability assumption, the space of smooth Jacobi fields has constant dimension for $s \in \mathcal{M}$.

Example (Jacobi fields for unions of planes). For a round S^2, the eigenspace $\ker(\Delta_{S^2} + 6)$ consists of the restrictions of trace-free quadratic polynomials on $\mathbb{R}^3$. The corresponding Jacobi fields integrate to deformations through phase-zero special Lagrangian 3-planes. The planes deform independently, so unions of great special Legendrian spheres are Jacobi-integrable.

2.3 A linear three-annulus estimate

In $\mathbb{R}_t \times \mathbb{R}^2_y$, for $K > 1$ put

$$U_r = \{(t, y) : |t| < r,\ r < |y| < Kr\}, \qquad \|v\|_r^2 = r^{-5} \int_{U_r} |v|^2.$$

Lemma 2.3. *Fix $3 < \gamma < 4$. For sufficiently large K and any $r > 0$, a nonzero t-independent closed and coclosed one-form v on $\mathbb{R} \times \{y \in \mathbb{R}^2 : r/K < |y| < K^2 r\}$ satisfies*

$$\|v\|_r^2 \geq K^3 \|v\|_{Kr}^2 \quad \Longrightarrow \quad \|v\|_{r/K}^2 > 2K^\gamma \|v\|_r^2.$$

Proof. The proof is modelled on the three-annulus arguments in [5, 17]. Write $v = c(y)\, dt + a(y)\, dy_1 + b(y)\, dy_2$. Closedness and coclosedness say that c is constant and $a - ib$ is holomorphic in $y_1 + iy_2$. Integrating in t gives

$$\|v\|_\rho^2 = 2\rho^{-4} \int_{\rho<|y|<K\rho} (|a|^2 + |b|^2 + |c|^2)\, dy.$$

The Laurent expansion of $a - ib$ decomposes the transverse part into orthogonal homogeneous terms of integer degrees k. A term of degree k has squared norm proportional to ρ^{2k-2}.

Let b_k^2 be the contribution of the k-th angular Fourier mode to the squared norm at scale r, with the contribution of $c\, dt$ included in b_0^2. Parseval's identity and dilation give

$$\|v\|_r^2 = \sum_k b_k^2, \qquad \|v\|_{Kr}^2 = \sum_k K^{2k-2} b_k^2, \qquad \|v\|_{r/K}^2 = \sum_k K^{2-2k} b_k^2.$$

The hypothesis gives

$$\sum_{k \geq 0} b_k^2 \leq K^2 \|v\|_{Kr}^2 \leq K^{-1} \|v\|_r^2.$$

Hence

$$\|v\|_{r/K}^2 \geq K^4 \sum_{k \leq -1} b_k^2 \geq K^4 (1 - K^{-1}) \|v\|_r^2 > 2K^\gamma \|v\|_r^2$$

for sufficiently large K, since $\gamma < 4$. □

2.4 Classification of transverse limits with planar ends

We recall the classification of transverse special Lagrangians with planar ends and the asymptotic coefficients of Lawlor necks [9, 10].

Lemma 2.4. *Let T be a boundaryless calibrated two-current in $\mathbb{C}^2$ with quadratic mass growth. Suppose that outside a compact set, T consists of two multiplicity-one normal graphs V_i over affine special Lagrangian planes Q_i, $i = 1, 2$. Assume that the planes have distinct directions and $|V_i(x)| \to 0$ as $|x| \to \infty$. Then $T = Q_1 \cup Q_2$, or T is a Lawlor neck asymptotic to $Q_1 \cup Q_2$.*

Proof. Write $z_j = x_j + iy_j$. Upon taking the hyperkähler rotation $w_1 = x_1 + ix_2$, $w_2 = y_1 - iy_2$, the Kähler form is $\operatorname{Re}(dz_1 \wedge dz_2)$, and the holomorphic volume form is $\omega - i \operatorname{Im}(dz_1 \wedge dz_2)$. The current is therefore a positive complex curve. In the projective compactification each decaying graphical end meets the line at infinity at one point with intersection multiplicity one. The graph description gives finite projective area there, so the removable-singularity theorem for analytic curves closes the end. The resulting projective curve has degree two and, by Chow's theorem, is the union of two lines or an irreducible conic. In suitable complex coordinates Z_1, Z_2, the conic has equation $Z_1 Z_2 = a + ib$, with $a, b \in \mathbb{R}$, and is a Lawlor neck. □

2.5 A criterion for Lawlor neck extraction

Fix a compact family of pairs of affine special Lagrangian 3-planes $\Pi_\pm = \mathbb{R}\vec{v} + z_\pm + Q_\pm$, where $\vec{v}$ is a unit vector, $z_\pm \perp \vec{v}$, and $Q_-, Q_+ \subset (\mathbb{C}\vec{v})^\perp$ are uniformly transverse. Fix $K > 1$ and $C_1 > 1 + \max_\pm |z_\pm|$ uniformly over this family. Write $x = t\vec{v} + z_\pm + y$, where $y \in Q_\pm$, and set $r = |y|$.

Lemma 2.5. *For any fixed ball B centred at 0, there is $C > 1$ such that, for every $\varepsilon > 0$, there are $C_2(\varepsilon) > C_1 + 2K$ and $\varepsilon_1 > 0$ with the following property. Let T be a closed special Lagrangian integral current in $B_{C_2(\varepsilon)} \supset \overline{B}$. On $\{r > C_1,\ |t| < r/(8K)\} \cap B_{C_2(\varepsilon)}$, assume T can be written as a multiplicity-one normal graph V over the affine planes, with $|V| + r|\nabla V| + r^2|\nabla^2 V| \le C_1/r$. If*

$$\|T\|(B_{C_2(\varepsilon)}) \le C_1 C_2(\varepsilon)^3, \qquad \int_{B_{C_2(\varepsilon)}} |\vec{v}^\perp|^2 \, d\|T\| \le \varepsilon_1,$$

$$\int_{B_{C_1+2K}} \operatorname{dist}(x, \Pi_- \cup \Pi_+)^2 \, d\|T\|(x) \ge C_1^{-1},$$

then T in B is a multiplicity-one C^2 graph of norm at most ε over a Lawlor neck times $\mathbb{R}$, with $C^{-1} \le |a + ib| \le C$.

Proof. We suppose for contradiction that the assertion fails for some fixed B, ε. Choose counterexamples T_i on B_i, with $\int_{B_i} |\vec{v}_i^\perp|^2 \, d\|T_i\| \to 0$. By the compactness of minimal surfaces, after passing to a subsequence for which the affine planes converge, T_i converges locally to T_∞ as currents and varifolds. The bound on $\vec{v}^\perp$ makes T_∞ translation invariant along $\vec{v}$, so $T_\infty = \mathbb{R}\vec{v} \times S$.

The lower distance bound passes to the limit. The exterior graphical bound gives smooth convergence on compact subsets there and $|V_\infty| \le Cr^{-1}$. Thus S has two multiplicity-one ends decaying to the prescribed affine planes, and the mass bound gives quadratic mass growth.

The Lagrangian condition makes the $J\vec{v}$ coordinate constant on each component of S, which is calibrated by $\iota_{\vec{v}} \operatorname{Re} \Omega$ in a copy of $\mathbb{C}^2$. A component with one end is its affine asymptote by the monotonicity formula. Lemma 2.4 therefore shows that S is either the union of

the two affine planes or a Lawlor neck. Any additional compactly supported component is zero by the monotonicity formula. The planar alternative contradicts the positive distance integral.

Evaluating $Z_1 Z_2 = a + ib$ on a fixed exterior graphical region bounds $|a+ib|$ above. The lower distance bound keeps it uniformly away from zero, since necks with $a+ib \to 0$ converge to the two affine planes.

Allard regularity and interior estimates now give C^2 graphical convergence to the product in B, a contradiction. □

Remark. With only one affine plane and without the lower distance bound, the conclusion is a normal graph over that plane with C^2 norm at most ε. The same conclusions hold if the normal graphs V are taken over submanifolds that are themselves C^4 graphs of norm at most ε_1 over the affine planes in $B_{C_2(\varepsilon)}$. These submanifolds have the same affine limits, so the proof is unchanged.

3 Graphical extension and Lawlor neck extraction

Throughout this section and Section 4 we work at unit scale and write $D = D(1)$ and $E = E(1)$. Our initial smallness conditions are

$$\begin{gathered}\int_{L\cap(B_4\setminus\overline{B}_{1/4})} \operatorname{dist}(x,\mathbf{C}_0)^2\, d\|L\|(x) + \int_{\mathbf{C}_0\cap(B_4\setminus\overline{B}_{1/4})} \operatorname{dist}(x,\operatorname{spt} L)^2\, d\|\mathbf{C}_0\|(x) < \delta^2, \\ \|L\|(B_4) < \|\mathbf{C}_0\|(B_4) + \delta.\end{gathered} \tag{3.1}$$

We use δ for initial closeness and ρ_0 for the smallness threshold on stopping radii. For sufficiently small δ, fix a *best approximating cone* $\mathbf{C}_s$, with $s \in \operatorname{int}\mathcal{M}$, attaining the infimum defining D.

On small balls in the regular part of $\mathbf{C}_0$, sufficiently small L^2 height and mass ratios close to one give multiplicity-one normal graphs by Allard regularity. Interior estimates then control the C^1 norm by the L^2 height on slightly larger balls. Conditions (3.1) ensure this on fixed compact regions of $\mathbf{C}_0\cap(B_4\setminus\overline{B}_{1/4})$ away from the singular rays. The task of this section is to extend these graphs towards the singular rays and describe L in their neighbourhoods.

3.1 Reduction to the concentrated case

We show that small E/D^2 forces concentration of the normalized height near the rays through the double points of the link.

Let $\pi : S^5 \to \mathbb{CP}^2$ be the Hopf projection. Near each original double point, transversality gives a unique nearby intersection of the two projected branches of Λ_s, depending smoothly on s. Let q_-, q_+ be the corresponding points on the two local sheets of Λ_s. Choose the parametrizations F_s so that their preimages in Σ are fixed, and denote this finite set by $\mathcal{P}$. Let $\mathcal{R}_s$ be the union of the rays through these points.

Lemma 3.1. *For every $\varepsilon > 0$ and $0 < \tau < 1/8$, there are $\delta, \varepsilon_1 > 0$, independent of s, such that, if (3.1) holds, $\mathbf{C}_s$ is a best approximating cone with $s \in \operatorname{int}\mathcal{M}$, $D > 0$ and $E \le \varepsilon_1 D^2$, then L is given by normal graphs V over the part of $\mathbf{C}_s$ with $1/2 < |x| < 3$ and $\operatorname{dist}(x,\mathcal{R}_s) > \tau/2$, and*

$$\int_{\mathbf{C}_s\cap\{1/2<|x|<3,\ \operatorname{dist}(x,\mathcal{R}_s)>\tau\}} |V|^2 \le \varepsilon D^2.$$

Proof. Suppose the assertion fails for a sequence L_i with $\delta_i \downarrow 0$, $E_i/D_i^2 \to 0$, and best approximating cones $\mathbf{C}_{s_i}$. Compactness and (3.1) give a stationary integral varifold limit supported on $\mathbf{C}_0 \cap (B_4 \setminus \overline{B}_{1/4})$. The second height integral in (3.1) and the monotonicity formula ensure that every sheet is present. Constancy, monotonicity and the mass bound give multiplicity one, so $L_i \to \mathbf{C}_0$ as varifolds on $B_4 \setminus \overline{B}_{1/4}$. Allard regularity gives the normal graphs V_i on this region and on every other compact regular subregion of $B_4 \setminus \overline{B}_{1/4}$. For a sufficiently small C^1 normal graph $Y(R,p) = RF_s(p) + V(R,p)$, the vector $V - R\partial_R V = Y - R\partial_R Y$ is normal to the cone, and its projection onto the normal space of the graph is $Y^{\perp}$. Hence

$$|V - R\partial_R V| = R^2 \left| \partial_R \left(\frac{V}{R} \right) \right| \leq C|Y^{\perp}|. \tag{3.2}$$

By the interior estimates, a subsequence of V_i/D_i converges smoothly on compact regular regions to a Jacobi field V_∞. The inequality in (3.2) and $E_i/D_i^2 \to 0$ give $R\partial_R V_\infty = V_\infty$. Thus $V_\infty = J\nabla(R^2\phi)$. Fatou's lemma and the contradiction hypothesis imply $V_\infty \in L^2$. Lemma 2.2 now removes the punctures and gives $\phi \in \ker(\Delta_\Sigma + 6)$ on the compact domain Σ.

Since each $\mathbf{C}_{s_i}$ is a best approximating cone, integrability implies that V_∞ is orthogonal to every smooth degree-one Jacobi field in the annular L^2 inner product. (The contribution of the L^2 pairing from the concentrated region is negligible because, with

$$W_{\tau'} = \{x \in B_4 \setminus \overline{B}_{1/4} : \operatorname{dist}(x, \mathcal{R}_0) < \tau'\},$$

the Cauchy–Schwarz inequality bounds it by

$$\frac{C}{D_i} \int_{W_{\tau'}} \operatorname{dist}(x, \mathbf{C}_{s_i})\, d\|L_i\|(x) \leq C\|L_i\|(W_{\tau'})^{1/2}.$$

By varifold convergence, the right-hand side tends to zero as $i \to \infty$ and then $\tau' \downarrow 0$.) Since V_∞ is a smooth degree-one Jacobi field, it is orthogonal to itself. Hence $V_\infty = 0$, contradicting the lower height bound in the contradiction hypothesis. □

3.2 Qualitative graphical extension

Our goal is to extend the normal graphs closer to $\mathcal{R}_s$. We prove a local criterion under approximate translation invariance.

For a unit vector $\vec{v} \in \mathbb{C}^3$, let $Q_-, Q_+ \subset (\mathbb{C}\vec{v})^{\perp}$ be transverse real two-planes calibrated by $\iota_{\vec{v}} \operatorname{Re}\Omega$. Fix a compact family $\mathcal{F}$ of such triples $(\vec{v}, Q_-, Q_+)$. For $z_-, z_+ \in \mathbb{R}J\vec{v}$, set

$$P_\pm = \mathbb{R}\vec{v} + z_\pm + Q_\pm, \qquad \ell_\pm = \mathbb{R}\vec{v} + z_\pm.$$

Thus P_-, P_+ are affine special Lagrangian 3-planes. They intersect along $\ell_- = \ell_+$ if $z_- = z_+$, and are disjoint otherwise. On each plane write $x = t\vec{v} + z_\pm + y$, with $y \in Q_\pm$, and recall $U_1 = \{(t,y) : |t| < 1,\ 1 < |y| < K\}$ (cf. Lemma 2.3).

Lemma 3.2. *Fix a sufficiently large K, and let $C_1 > 1$ and $\eta > 0$. There is $\varepsilon_0 > 0$ with the following property. Let $(\vec{v}, Q_-, Q_+) \in \mathcal{F}$, assume $|z_-|, |z_+| \leq C_1$, and define $P_\pm, \ell_\pm$ as above. Let T be a closed special Lagrangian integral current in B_K with $\|T\|(B_K) \leq C_1$. Suppose that outside the unit tubes about ℓ_-, ℓ_+, T is given by multiplicity-one normal graphs $V_\pm$ over $P_\pm$. Assume these fields extend as special Lagrangian graphs with C^2 norm at most one to $\{|t| < 2K,\ 1/2 < |y| < 2K\}$ on each plane, and satisfy*

$$\max_\pm \|V_\pm\|_{L^2(U_1)} \leq \varepsilon_0, \qquad \int_{B_K} |\vec{v}^{\perp}|^2\, d\|T\| \leq \varepsilon_0^2.$$

Then the graphical representation extends to B_4 outside the tubes of radius $1/(2K)$ about ℓ_-, ℓ_+, with $\max_\pm \|V_\pm\|_{C^2} \leq \eta$.

Proof. We follow the compactness argument of [3, Proposition 3.2]. Suppose the conclusion fails as $\varepsilon_0 \downarrow 0$. The parameter and mass bounds give a subsequence with convergent model parameters and $T_i \to T_\infty$ as currents and varifolds.

The uniform C^2 bound and interior elliptic estimates give C^2 convergence of the graphical fields on compact subsets of $\{|t| < 2K,\ 1/2 < |y| < 2K\}$. Their limits vanish on U_1 by the L^2 smallness, and hence on this connected domain by unique continuation. The portions of $P_\pm \cap B_K$ outside the unit tubes are compactly contained in this domain, so the fields tend to zero there in C^2. Thus T_∞ is closed and calibrated, and agrees with $[P_-] + [P_+]$ in B_K outside the unit tubes about the two axes.

The vanishing of $\int_{B_K} |\vec{v}_i^{\perp}|^2 \, d\|T_i\|$ implies that $\vec{v}$ is tangent to T_∞ almost everywhere in B_K. Hence T_∞ is translation invariant along the $\vec{v}$ direction. The singular set has Hausdorff dimension at most one and does not disconnect the 3-planes. By real analyticity, each nonempty $P_\pm \cap B_{K/2}$ is a multiplicity-one component of T_∞. Since T_∞ agrees with the two planes outside the unit tubes, translation invariance implies that any remaining component meeting $B_{K/2}$ has a compactly supported closed calibrated slice transverse to $\vec{v}$. This slice is zero by the monotonicity formula. Thus $T_\infty = [P_-] + [P_+]$ in $B_{K/2}$.

On each $P_\pm$, the part of $\overline{B}_4$ at distance at least $1/(2K)$ from the axes is a compact subset of the multiplicity-one regular set of the limit. Allard regularity [12, Section 3] and interior estimates give the required C^2 bounds for graphs of T_i over $P_{\pm,i}$. Uniqueness of normal projection identifies them with the original graphs on overlaps, contradicting the choice of sequence. The same argument applies when the affine planes are replaced by sheets converging to them in C^2. □

Remark. For a single model plane, the same argument gives extension over its intersection with B_4, with the same bound.

3.3 The nonlinear three-annulus estimate

We return to L and its comparison cone $\mathbf{C}_s$. Given $q \in \mathcal{R}_s$ with $2/3 \le |q| \le 7/3$, write $V(t, y)$ for the normal vector field representing L as a graph over the corresponding smooth branch of $\mathbf{C}_s$, in coordinates centred at q. Here t is arclength along the ray, with $t = 0$ at q, and the transverse coordinates y are orthonormal there. Recall U_r from Lemma 2.3 and put

$$\widetilde{U}_r = \{(t, y) : |t| < 2Kr,\ r/2 < |y| < 2Kr\}.$$

Define the norms

$$\|V\|_r^2 = r^{-5} \int_{U_r} |V|^2, \qquad \|V\|_{C_r^2} = \sum_{j=0}^{2} r^j \|\nabla^j V\|_{L^\infty(\widetilde{U}_r)}.$$

The integral uses the induced measure on this cone branch.

Set

$$\mathcal{E}(q, r) = r^{-3} \int_{L \cap B_{Cr}(q)} |x^{\perp}|^2 \, d\|L\|,$$

where C is fixed and large enough that $B_{Cr}(q)$ contains the graphs over $\widetilde{U}_{r/K} \cup \widetilde{U}_r \cup \widetilde{U}_{Kr}$. For sufficiently small r, monotonicity gives $\mathcal{E}(q, r) \le C r^{-3} E$.

Proposition 3.3. *Fix $3 < \gamma < 4$ and choose K sufficiently large as in Lemma 2.3. There is $\eta > 0$ such that, for sufficiently small $r > 0$, if the graph is defined on $\widetilde{U}_{r/K} \cup \widetilde{U}_r \cup \widetilde{U}_{Kr}$, then*

$$\|V\|_r^2 \ge K^3 \|V\|_{Kr}^2 \quad \Longrightarrow \quad \|V\|_{r/K}^2 \ge 2K^\gamma \|V\|_r^2,$$

provided $\rho^{-1}\|V\|_{C_\rho^2} \le \eta$ for $\rho \in \{r/K, r, Kr\}$ and $\mathcal{E}(q, r) \le \eta \|V\|_r^2$. Constants are uniform over the fixed compact model family.

Proof. Suppose the assertion fails at centres q_i as η and the local radius tend to zero. Set

$$v_i(t,y) = \frac{V_i(r_i t, r_i y)}{r_i \|V_i\|_{r_i}}.$$

The hypothesis and the failure of the conclusion give

$$\|v_i\|_K^2 \le K^{-3}, \qquad \|v_i\|_1^2 = 1, \qquad \|v_i\|_{1/K}^2 < 2K^\gamma.$$

After rescaling, the corresponding branch of the comparison cone converges smoothly to a 3-plane. The height bounds and (3.2) give

$$\int_{U_\sigma} |\partial_t v_i|^2 \le C\left(r_i^2 + \frac{\mathcal{E}(q_i, r_i)}{\|V_i\|_{r_i}^2}\right) \longrightarrow 0, \qquad \sigma \in \{K^{-1}, 1, K\}.$$

By interior regularity of the special Lagrangian graph equation, the three height bounds give local smooth convergence of the normalized fields to v. The derivative estimate makes v independent of t and gives strong L^2 convergence on U_1, hence $\|v\|_1^2 = 1$.

Lower semicontinuity gives

$$\|v\|_K^2 \le K^{-3}, \qquad \|v\|_{1/K}^2 \le 2K^\gamma.$$

The one-form associated with v is closed and coclosed and has the same norms. Since $\|v\|_1^2 = 1$, Lemma 2.3 gives $\|v\|_{1/K}^2 > 2K^\gamma$, contradicting the second bound. □

3.4 Graphical extension up to the stopping radii

We shall extend the normal graphs towards $\mathcal{R}_s$ by successively replacing the inner radius r in the domains U_r by r/K. For $R_0 \in [2/3, 7/3]$, we consider the domains $U_r \subset \widetilde{U}_r$ centred at $R_0 q_\pm$ on each local sheet of $\mathbf{C}_s$.

Set

$$D_{R_0}(r)^2 = r^{-5} \int_{L \cap \{|t - R_0| < r,\ r < |y| < Kr\}} \operatorname{dist}(x, \mathbf{C}_s)^2 \, d\|L\|.$$

In this integral, (t, y) are the coordinates of the normal projection onto each sheet, with t measured from the origin along its ray; the integral is summed over the two sheets.

Fix $3 < \gamma < 4$ and choose K sufficiently large. Choose η and ε_0 sufficiently small as in Proposition 3.3 and Lemma 3.2, respectively. Then choose r_0 sufficiently small depending on these constants and the fixed family of nearby cones. Finally, choose δ in (3.1) sufficiently small that Allard regularity supplies the initial graphs with rescaled C^2 norm at most η and $D_{R_0}(r_0)^2, D_{R_0}(Kr_0)^2 < \varepsilon_0^2$. Put $r_j = K^{-j} r_0$ for $j \ge 0$. At step j, the graphs are already defined on $\widetilde{U}_{r_j} \cup \widetilde{U}_{Kr_j}$. Apply the following stopping criteria in order:

1. *Energy criterion.* Stop if $\mathcal{E}(R_0 q_\pm, r_j) \ge \varepsilon_0^2$ for at least one of the two signs.
2. *Height criterion.* Stop if $D_{R_0}(r_j)^2 \ge \varepsilon_0^2$.

If neither holds, extend the graphs to $\widetilde{U}_{r_{j+1}}$ and repeat with $j+1$. At centres where neither criterion is ever met, continue at every scale.

Proposition 3.4 (Inductive graphical extension)**.** *With these choices, the construction is well defined at every centre. At every step j, L is given by normal graphs over the corresponding domains of $\mathbf{C}_s$, with*

$$r_j^{-1} \|V\|_{C^2_{r_j}} \le \eta$$

on both sheets.

Proof. The initial graphs and their bounds follow from the choice of δ. Suppose the graphs have been constructed down to r_j and neither stopping criterion holds. After rescaling by r_j^{-1}, the fields are defined on $\widetilde{U}_1$ with C^2 norm at most $\eta < 1$. Uniform transversality makes normal height comparable to distance from $\mathbf{C}_s$ on U_{r_j}, so on each sheet

$$\|V\|_{r_j} \le CD_{R_0}(r_j) < C\varepsilon_0.$$

This verifies the L^2 hypothesis in Lemma 3.2.

At either centre $q = R_0 q_\pm$, put $\vec{v} = q/|q|$. The monotonicity formula gives the required rescaled mass bound, and $|q|\vec{v}^\perp = x^\perp - (x-q)^\perp$ gives

$$r_j^{-3} \int_{L \cap B_{Cr_j}(q)} |\vec{v}^\perp|^2 \, d\|L\| \le C(\mathcal{E}(q, r_j) + r_j^2) \le C(\varepsilon_0^2 + r_0^2).$$

Up to a negligible error of order $O(r_j)$ in the rescaled C^2 norm, we can replace each local sheet of $\mathbf{C}_s$ by its affine tangent plane at $R_0 q_\pm$. If both sheets meet $B_{Cr_j}(q)$, their centres are $O(r_j)$ apart, so their radial directions differ by $O(r_j)$. Hence, up to SU(3) rotations of order $O(r_j)$, we can assume that both radial axes are parallel to $\vec{v}$. Since the centres lie in $\mathbb{C}\vec{v}$, the separation of these axes lies in $\mathbb{R}J\vec{v}$. This gives the planes required in Lemma 3.2. Apply Lemma 3.2 with output bound $\eta/(2K)$ (using its single-plane version if only one sheet meets the ball). Since

$$\widetilde{U}_{r_{j+1}} = \{|t| < 2r_j,\ r_j/(2K) < |y| < 2r_j\} \subset B_{4r_j} \cap \{|y| > r_j/(2K)\},$$

Lemma 3.2 supplies the graphs near $\overline{\widetilde{U}_{r_{j+1}}}$ with rescaled C^2 norm at most $\eta/2$ at scale r_{j+1}. Returning to the comparison cone adds $O(Kr_j)$, absorbed by the choice of r_0, and gives $r_{j+1}^{-1}\|V\|_{C^2_{r_{j+1}}} \le \eta$. This completes the induction. □

At a stopping radius r_j, call $B_{Cr_j}(R_0 q_\pm)$ the terminal balls.

3.5 A lower bound for a stopping radius

For the normal vector fields from the stopping construction, put

$$Q = \sup_{V,j} r_j^\gamma \|V\|_{r_j}^2,$$

where r_j ranges over all scales at which V is defined by the stopping construction, including stopping radii. Consequently, $D_R(r)^2 \le CQr^{-\gamma}$ for every R and every scale r at which the graphs are defined.

Proposition 3.5. *There are $\delta, \varepsilon_1 > 0$ and $C > 1$ with the following property. Suppose (3.1) holds, $D > 0$ and $E \le \varepsilon_1 D^2$. Then $Q \ge C^{-1}D^2$. At some regular radius $R_0 \in [2/3, 7/3]$, the construction stops by the height criterion at a radius ρ with*

$$C^{-1}Q \le \rho^\gamma \le CQ.$$

Every stopping radius satisfying the height criterion is at most $C\rho$.

Proof. Step 1: bound Q from below. Put $f(x) = \operatorname{dist}(x, \mathbf{C}_s)$. Choose a fixed Lipschitz function χ, supported in $[1/4, 4]$, and a constant $c_0 > 0$ such that

$$R\chi'(R) + 5\chi(R) = 1 - c_0 \mathbf{1}_{(1,2)}(R), \qquad 1/4 < R < 4.$$

Since f^2 is homogeneous of degree two and $|D(f^2)| \le 2f$, stationarity tested with $\chi(R)f^2 x$ gives

$$D^2 \le c_0 \int_{L\cap\{1<R<2\}} f^2 \, d\|L\| + CD\sqrt{E}.$$

Fix a sufficiently small $0 < \tau < 1/8$. Lemma 3.1 and $E \le \varepsilon_1 D^2$ imply, after decreasing δ and ε_1, that

$$\int_{L\cap\{1<R<2,\ \mathrm{dist}(x,\mathcal{R}_s)<\tau\}} f^2 \, d\|L\| \ge C^{-1}D^2.$$

Cover the set above, up to a subset of $\mathcal{R}_s$ of zero $\|L\|$-measure, by the images of U_{r_j} under the normal graph parametrizations and by terminal balls. Choosing $C^{-1}r_j$-separated centres along each ray gives at most C/r_j regions at scale r_j, with bounded overlap of the associated balls. For r_0 small, all these balls lie in $B_4 \setminus \overline{B}_{1/4}$.

Each graphical region contributes at most $CQr_j^{5-\gamma}$ to the integral of f^2. For a terminal ball $B_{Cr_j}(q)$, monotonicity and $q \in \mathbf{C}_s$ give

$$\int_{B_{Cr_j}(q)} f^2 \, d\|L\| \le Cr_j^2 \|L\|(B_{Cr_j}(q)) \le Cr_j^5.$$

At a height stopping radius, $r_j^\gamma \le CQ$, so this is also bounded by $CQr_j^{5-\gamma}$. For balls satisfying the energy criterion, bounded overlap gives a total contribution at most $Cr_j^2 E$ at each scale. Summing over the scales $r = r_0, K^{-1}r_0, \dots$ and using $\gamma < 4$ gives

$$D^2 \le CQ \sum_{\text{scales } r} r^{4-\gamma} + CE \sum_{\text{scales } r} r^2 \le C(Q+E).$$

Taking ε_1 sufficiently small and absorbing the E term, we obtain $Q \ge C^{-1}D^2$.

Step 2: select a height stopping radius. Choose a centre $q \in \mathcal{R}_s$, with $R_0 = |q|$, and a radius ρ at which the corresponding field V is defined, such that

$$Q/2 < \rho^\gamma \|V\|_\rho^2 \le C\rho^\gamma,$$

where the upper bound follows from the uniform graphical estimate. (By a small perturbation of the centre, we may assume that the slice at R_0 is regular.) Since $Q \ge C^{-1}D^2$, Lemma 3.1 gives $r_0^\gamma \|V\|_{r_0}^2 < Q/2$ at every centre once δ and ε_1 are sufficiently small. Thus $\rho < r_0$, so the graphs are also defined at scale $K\rho$. Since $\gamma > 3$ and K is sufficiently large, the definition of Q gives

$$\|V\|_\rho^2 > \tfrac{1}{2}K^\gamma \|V\|_{K\rho}^2 \ge K^3 \|V\|_{K\rho}^2.$$

Moreover, $E \le \varepsilon_1 D^2$ and $Q \ge C^{-1}D^2$ give

$$\frac{E}{\rho^3 \|V\|_\rho^2} \le C\frac{E}{Q}\rho^{\gamma-3} \le C\varepsilon_1 \rho^{\gamma-3}.$$

This verifies the normalized energy bound in Proposition 3.3 for sufficiently small ε_1. Also, $E/\rho^3 \le C\varepsilon_1 Q/\rho^3 \le C\varepsilon_1 \rho^{\gamma-3}$, so the energy criterion fails at every centre and every radius at least ρ.

If the height criterion failed at R_0 and ρ, the construction would reach ρ/K, and Proposition 3.3 would give

$$(\rho/K)^\gamma \|V\|_{\rho/K}^2 \ge 2\rho^\gamma \|V\|_\rho^2 > Q,$$

contrary to the definition of Q. Thus ρ is a height stopping radius. At any radius r satisfying the height criterion,

$$\varepsilon_0^2 r^\gamma \le r^\gamma D_R(r)^2 \le CQ.$$

Together with $Q \le C\rho^\gamma$, this proves $C^{-1}Q \le \rho^\gamma \le CQ$ and $r \le C\rho$. □

3.6 Estimates on the graphical region

The Legendrian neighbourhood construction, applied along F_s, gives local diffeomorphisms Ψ_s from a neighbourhood of the zero section of $T^*\Sigma \times \mathbb{R}$ into S^5, depending smoothly on s, such that

$$\Psi_s(p,0,0) = F_s(p), \qquad \Psi_s^*\lambda = 2\left(dz - \sum_j \xi_j\, dp^j\right).$$

Here (p^j, ξ_j) are local cotangent coordinates and z is the coordinate on the $\mathbb{R}$-factor. Choose these coordinates so that the infinitesimal displacement in the fibre directions is $J\xi^\sharp + 2zJF_s$. Let $\Gamma_R = (\lambda_{0,R})_\#\langle L, |x|, R\rangle$ be the radial slice normalized to S^5, and write its graphical part as $\Psi_s(p, \alpha_R(p), u_R(p))$, where α_R is a one-form and u_R is a real-valued function. The graphical estimates of Proposition 3.4 apply in these coordinates.

Recall the stopping radius ρ from Proposition 3.5. Fix $1 < p < 2$ and put $\rho_* = \rho^p$. Assume $E \le \rho_*^5$ and take ρ sufficiently small. The construction then reaches ρ_* wherever the height criterion has not been met. Fix any radial value $R \in [2/3, 7/3]$ and a puncture $p_\pm \in \mathcal{P}$ corresponding to $q_\pm$. Let r_j be the stopping radius at $Rq_\pm$, taken to be zero if the construction never stops, and put

$$\sigma = \max\{r_j, \rho_*\}.$$

Let $\bar{u}_R$ be the circular average of u_R around $p_\pm$. The norms C^k_r below are computed after rescaling each annulus to unit size.

Proposition 3.6 (Estimates for the potential)**.** *There are $\delta, \rho_0, C > 0$ such that, if (3.1) holds, $0 < \rho < \rho_0$ and $E \le \rho_*^5$, then $\|u_R\|_{C^2(\Sigma_{r_0})} \le C\rho^2$, where $\Sigma_{r_0} = \Sigma \setminus \bigcup_{p\in\mathcal{P}} B^\Sigma_{r_0}(p)$. In geodesic polar coordinates (r,θ) about $p_\pm$, for $\sigma < r < r_0$,*

$$|u_R(r,\theta)| \le C\rho^2(1 + \log(r_0/r)), \qquad \|u_R - \bar{u}_R\|_{C^2_r} \le C\rho^2,$$
$$|du_R| + r|D^2 u_R| \le C\left(\frac{\sigma^2}{r} + \rho^2\right). \tag{3.3}$$

For every $\varepsilon > 0$, there is $C(\varepsilon) > 1$ such that, after decreasing δ, ρ_0 if necessary, there is $a \in \mathbb{R}$ with $|a| \le C\sigma^2$ such that

$$|r\bar{u}_R'(r) - a| \le \varepsilon\sigma^2 + C\rho^2 r, \qquad C(\varepsilon)\sigma < r < r_0. \tag{3.4}$$

The constants are uniform in $R \in [2/3, 7/3]$ and independent of ρ and σ. We write $C(\varepsilon)$ for constants that may depend on ε, reserving C for constants independent of it.

Proof. Step 1: estimate the radial error. Write $Y_R(p) = \Psi_s(p, \alpha_R(p), u_R(p))$ and let d denote differentiation on Σ. Under the parametrization $(R,p) \mapsto RY_R(p)$ of L, the Liouville form pulls back to

$$2R^2(\partial_R u_R\, dR + du_R - \alpha_R).$$

Its closedness on L gives $d\alpha_R = 0$ and $R\partial_R\alpha_R = 2(du_R - \alpha_R)$. On Γ_R,

$$Y_R^*\lambda = 2(du_R - \alpha_R). \tag{3.5}$$

The restriction of λ to the three-dimensional regular part of L is harmonic and satisfies $|\lambda| = |x^\perp|$. Wherever Proposition 3.4 supplies the graphs, interior estimates on U_r give

$$r^j\|\nabla^j\lambda\|_{C^1_r} \le Cr^{-3/2}\sqrt{E}, \qquad j = 0, 1. \tag{3.6}$$

Since $2/3 \le R \le 7/3$, the pullback formula and the identity for $\partial_R \alpha_R$ give

$$\|\partial_R u_R\|_{C^2_r} + \|\partial_R \alpha_R\|_{C^2_r} + \|\partial_R Y_R\|_{C^2_r} \le C r^{-3/2}\sqrt{E}.$$

To derive an equation for u_R on each radial slice, use (3.5) to substitute $\alpha_R = du_R - \frac{1}{2} Y_R^* \lambda$ in the special Lagrangian equation for L, and collect the Liouville and radial terms in an error term f, controlled by (3.6). Let $D^2 u_R$ denote the covariant Hessian for the metric induced by F_s. The special Legendrian operator for the graph $\Psi_s(p, du_R, u_R)$ is at most quadratic in $D^2 u_R$ and at $u_R = du_R = 0$ equals $\operatorname{tr} D^2 u_R$. Its linearization is $\operatorname{tr} D^2 u_R + 6u_R$, so the special Lagrangian equation has the schematic form

$$\begin{aligned}\operatorname{tr} D^2 u_R + 6u_R = {} & (u_R, du_R)\#(u_R, du_R) + (u_R, du_R)\# D^2 u_R \\ & + (u_R, du_R)\# D^2 u_R \# D^2 u_R + f.\end{aligned}$$

Here $\#$ denotes sums of contractions with smooth coefficients depending on (p, u_R, du_R); these include the curvature of Λ_s and the expression of Ω in the contact coordinates. The graphical bounds in Proposition 3.4 give

$$\|u_R\|_{C^1_r} + \|du_R\|_{C^1_r} \le Cr, \qquad \|D^2 u_R\|_{C^1_r} \le C.$$

In geodesic normal coordinates about the puncture in Σ, absorb the nonlinear terms involving $D^2 u_R$ into a^{ij} to obtain

$$a^{ij}\partial_i \partial_j u_R + b^i \partial_i u_R + c u_R = f. \tag{3.7}$$

Here c is uniformly bounded, and $E \le \rho_*^5 \le \sigma^5$ gives

$$\begin{aligned}\|f\|_{C^1_r} &\le C r^{-5/2}\sqrt{E} \le C(\sigma/r)^{5/2}, \\ \|a^{ij} - \delta^{ij}\|_{C^1_r} + \|b^i\|_{C^1_r} &\le Cr + C r^{-3/2}\sqrt{E} \le Cr.\end{aligned} \tag{3.8}$$

Step 2: local estimates in terms of the exterior norm. Set $M = \sup_{2/3 \le R \le 7/3} \|u_R\|_{C^2(\Sigma_{r_0})}$, and let $\Delta_0 = \partial_1^2 + \partial_2^2$ be the Euclidean Laplacian in the normal coordinates. The graphical bounds and (3.5) give $\|u_R - \bar{u}_R\|_{C^2_\sigma} + |\sigma \bar{u}_R'(\sigma)| \le C\sigma^2$, while the boundary data at $r = r_0$ are bounded by M. For $\sigma < r_0/4$, define the finite quantity

$$A = 1 + \sup_{\sigma < r < r_0} \frac{\|du_R\|_{C^1_r}}{\sigma^2/r + M/r_0}.$$

Integrating along radial lines from r_0 gives $|u_R| \le CA\{M + \sigma^2(1 + \log(r_0/r))\}$. Hence (3.7)–(3.8) give

$$|\Delta_0 u_R - f| \le C(r|D^2 u_R| + r|du_R| + |u_R|) \le CA\left(\frac{\sigma^2}{r} + \frac{M}{r_0}\right).$$

The circular average satisfies

$$r\bar{u}_R'(r) - \sigma \bar{u}_R'(\sigma) = \int_\sigma^r t\, \overline{\Delta_0 u_R}(t)\, dt. \tag{3.9}$$

In the variable $t = \log r$, the Dirichlet Green function for each Fourier mode $k \ne 0$ is bounded by $(2|k|)^{-1} e^{-|k||t-s|}$. The boundary data and f contribute $C(\sigma^2 + rM/r_0)$ and $C\sqrt{E}\, r^{-1/2} \le C\sigma^2$, respectively. Together with the bound for $\Delta_0 u_R - f$, this gives

$$|u_R - \bar{u}_R| \le C\left(\sigma^2 + \frac{r}{r_0} M\right) + CAr\{\sigma^2(1 + \log(r_0/r)) + M\}.$$

Equation (3.9) gives the same bound for $r|\bar{u}_R'|$, using $\int_\sigma^{r_0} t(\sigma/t)^{5/2}\,dt \le 2\sigma^2$. Interior elliptic estimates for (3.7) on slightly larger annuli give

$$A \le C + CA \sup_{\sigma<r<r_0} r(1+\log(r_0/r)) \le C + Cr_0 A,$$

with C independent of σ and small r_0. Fix r_0 independently of σ so that $Cr_0 \le 1/2$; absorbing the last term gives $A \le 2C$. Thus $\|u_R - \bar{u}_R\|_{C^2_r} \le C(M+\sigma^2)$ and $|du_R| + r|D^2u_R| \le C(\sigma^2/r + M)$.

Step 3: bound the exterior norm. Suppose the bound fails along a sequence with $\delta_i, \rho_i \to 0$ and $M_i/\rho_i^2 \to \infty$. Set $f_i = \operatorname{dist}(\cdot, \mathbf{C}_{s_i})$. By monotonicity, the L^2 norm of $x^\perp/M_i$ on the full annulus is $O(\sqrt{E_i}/M_i) = o(1)$. Since $\sigma_i \le C\rho_i$, Steps 1–2 and (3.2) give, after passing to a subsequence, smooth convergence of V_i/M_i on compact regular regions to a homogeneous Jacobi field $J\nabla(R^2 v)$ with $\|v\|_{C^2(\Sigma_{r_0})} = 1$. Step 2 gives $|dv| \le C$ near the punctures, so Lemma 2.2 extends v smoothly across them.

On $1 < R < 2$, the tube of radius $C\rho_i$ contributes $O(\rho_i^3) = o(M_i)$ to $\int f_i$, and Step 2 gives uniform integrability of f_i/M_i elsewhere. Together with the vanishing radial error above, these bounds give strong L^1 convergence of f_i/M_i to $|J\nabla(R^2 v)|$ on $B_4 \setminus \overline{B}_{1/4}$.

By integrability, choose cone variations $\mathbf{C}_{s_i(t)}$ whose normal velocities W_i at $t=0$ converge smoothly to $J\nabla(R^2 v)$. The minimising property of $\mathbf{C}_{s_i}$ gives

$$\begin{aligned} 0 &= \frac{d}{dt}\Big|_{t=0} \int_{L_i \cap (B_4 \setminus \overline{B}_{1/4})} \operatorname{dist}(x, \mathbf{C}_{s_i(t)})^2 \, d\|L_i\| \\ &= -2 \int_{L_i \cap (B_4 \setminus \overline{B}_{1/4})} \langle x - y_i(x), W_i(y_i(x)) \rangle \, d\|L_i\|, \end{aligned}$$

where $y_i(x) \in \mathbf{C}_{s_i}$ is a nearest point to x. After division by M_i, the integrand converges in C^0 on compact regular regions to $|J\nabla(R^2 v)|^2$ and is bounded by Cf_i/M_i near the singular rays. Strong L^1 convergence therefore gives, after cancelling the radial factor,

$$\int_\Sigma (4v^2 + |dv|^2)\, d\mathrm{Area} = 0.$$

This contradicts the normalization of v and proves $M \le C\rho^2$. Step 2 now gives the stated bounds for u_R and its derivatives.

Finally, take $a = C(\varepsilon)\sigma \bar{u}_R'(C(\varepsilon)\sigma)$ for $C(\varepsilon)$ sufficiently large. The derivative bound gives $|a| \le C\sigma^2$, since $p < 2$ and $\rho^2/\sigma \le \rho^{2-p} \ll 1$. Substituting (3.8) and the bounds just proved into (3.9) gives

$$\begin{aligned} |r\bar{u}_R'(r) - a| &\le \int_{C(\varepsilon)\sigma}^r t\, |\overline{\Delta_0 u_R}(t)|\, dt \\ &\le C\rho^2 r + \varepsilon\sigma^2, \end{aligned}$$

after choosing $C(\varepsilon)$ large and then ρ_0 small. □

3.7 Lawlor neck extraction

Lemma 3.7. *Fix $0 < \varepsilon < 1$. There are $C(\varepsilon) > 1$ and $\delta, \rho_0 > 0$ with the following property. Assume (3.1) holds, $0 < \rho < \rho_0$ and $E \le \rho_*^5$. For every regular $R_0 \in [2/3, 7/3]$ and every stopping radius $\sigma > C(\varepsilon)\rho_*$, L in $\bigcup_\pm B_{C(\varepsilon)\sigma}(R_0 q_\pm)$ is a multiplicity-one normal graph V over a translated product of a Lawlor neck and a line, with $\|V\|_{C^2_\sigma} \le \varepsilon\sigma$. On the normalized slice Γ_{R_0}, the logarithmic coefficients in (3.4) may be chosen as $\pm a$ on the corresponding $\pm$ ends, where*

$$C^{-1}\sigma^2 \le |a| \le C\sigma^2.$$

Here the constant C is independent of ε.

Proof. Step 1: extract a nontrivial neck. Since

$$\mathcal{E}(R_0 q_\pm, \sigma) \le C\sigma^{-3}E \le C\sigma^2 < \varepsilon_0^2,$$

the energy criterion fails. The height criterion gives

$$\int_{L\cap\{|t-R_0|<\sigma,\ \sigma<|y|<K\sigma\}} \operatorname{dist}(x,\mathbf{C}_s)^2\, d\|L\|(x) \ge \varepsilon_0^2\sigma^5.$$

Consequently $\int_{U_\sigma}|V|^2 \ge C^{-1}\varepsilon_0^2\sigma^5$ on at least one sheet. At the corresponding centre $q \in \{R_0q_-, R_0q_+\}$, Proposition 3.6 and interior estimates give

$$|V| + r|\nabla V| + r^2|\nabla^2 V| \le C\sigma^2/r$$

on the exterior graphical region $\sigma \le r \le C_2(\varepsilon)\sigma$, for $C_2(\varepsilon)$ as in Lemma 2.5 and sufficiently small ρ_0. After rescaling by σ^{-1}, this is the graphical bound in Lemma 2.5. The monotonicity formula gives

$$\|L\|(B_{C_2(\varepsilon)\sigma}(q)) \le C(C_2(\varepsilon)\sigma)^3.$$

Consider the affine tangent planes to the local sheets of $\mathbf{C}_s$ at $R_0q_\pm$. The centres satisfy $R_0|q_+ - q_-| \le C\sigma$. Otherwise the single-plane version in the remark after Lemma 2.5, applied at scale σ, would give a normal graph over the local sheet through q with $\|V\|_{C^2_\sigma} \le c\varepsilon_0\sigma$, contradicting the lower bound on $\int_{U_\sigma}|V|^2$ for sufficiently small c.

The radial directions of the two planes therefore differ by $O(\sigma)$. An $O(\sigma)$ special unitary rotation of one of the two planes about its centre makes both parallel to $\vec{v} = q/|q|$. In $B_{C_2(\varepsilon)\sigma}(q)$, the local sheets of $\mathbf{C}_s$ are then C^4_σ graphs of norm $O(C(\varepsilon)\sigma^2)$ over these planes. Since $q^\perp = x^\perp - (x-q)^\perp$, the energy and mass bounds give

$$\sigma^{-3}\int_{L\cap B_{C_2(\varepsilon)\sigma}(q)} |\vec{v}^{\,\perp}|^2\, d\|L\| \le C(\varepsilon)(\sigma^{-3}E + \sigma^2) \le C(\varepsilon)\sigma^2.$$

The lower bound on $\int_{U_\sigma}|V|^2$ verifies the remaining height hypothesis of Lemma 2.5 after rescaling by σ^{-1}, for suitable choices of C_1 and ρ_0. Lemma 2.5 therefore gives the claimed normal graph on a ball containing both terminal balls, with neck parameter of modulus comparable to σ^2.

Step 2: identify the coefficients. To identify the coefficients of u_{R_0}, work on Γ_{R_0} and apply the dilation λ_{0,R_0} to the Lawlor neck model. Choose the model coordinates so that the logarithmic term at the plus end is $a\log r$, and write its equation as $Z_1Z_2 = a + ib$. Then

$$\int_{|Z_1|=r} \lambda = 4\pi b.$$

Comparing this with the integral over the corresponding loop on Γ_{R_0} and using (3.6) gives

$$|b| \le C\varepsilon\sigma^2 + C\sigma^{-1/2}\sqrt{E} \le C\left(\varepsilon + \left(\frac{\rho_*}{\sigma}\right)^{5/2}\right)\sigma^2.$$

Choosing a sufficiently accurate approximation and a sufficiently large cutoff constant in $\sigma > C(\varepsilon)\rho_*$ gives $|a| \asymp \sigma^2$. The model potentials have logarithmic terms $\pm a\log r$ on the corresponding $\pm$ ends. The neck approximation at $r = C(\varepsilon)\sigma$ identifies their coefficients with those in (3.4) to accuracy $\varepsilon\sigma^2$. □

Flux interpretation of a. For the points $q_-, q_+ \in \Lambda_s$ corresponding to each double point of Λ, put

$$P = q_- q_-^* = q_+ q_+^*, \qquad \eta_H = \iota_{JHx}\iota_x \operatorname{Re}\Omega$$

for any trace-free Hermitian matrix H. On Γ_{R_0}, denote the graphical circles at $r = C(\varepsilon)\sigma$ by c_-, c_+, oriented as boundaries of the corresponding disks $B^{\Sigma}_{C(\varepsilon)\sigma}(p_\pm)$.

Claim 1. For every regular $R_0 \in [2/3, 7/3]$ and every such pair q_-, q_+, the sum of the fluxes through c_- and c_+ satisfies

$$\left| \int_{c_-} \eta_H + \int_{c_+} \eta_H \right| \le C\|H\|\sqrt{E}. \tag{3.10}$$

Moving either circle within its graphical annulus changes its flux by at most $C\|H\|\sqrt{E}$.

Proof. The three-dimensional graphical collars allow us to keep the circles $r = C(\varepsilon)\sigma$ fixed in the link coordinates as R varies over an interval I containing R_0, with $|I| \asymp \sigma$. Write $F_H(R)$ for the sum of their fluxes on Γ_R. Restrict each slice to the cutoff neighbourhood bounded by these circles. The corresponding undilated region $S \subset L$ lies in $\bigcup_\pm B_{C(\varepsilon)\sigma}(R_0 q_\pm)$, so $\|L\|(S) \le C(\varepsilon)\sigma^3$. The identity

$$d\eta_H = -3\iota_{JHx} \operatorname{Re}\Omega \tag{3.11}$$

and the special Lagrangian condition give

$$\int_I |F_H(R)|\, dR \le C\|H\| \int_S |x^\perp|\, d\|L\| \le C(\varepsilon)\|H\|\sigma^{3/2}\sqrt{E}.$$

On the circles, Cartan's formula and the bound for $\partial_R Y_R$ in Step 1 of Proposition 3.6 give

$$|F_H'(R)| \le C(\varepsilon)\|H\|\sigma\|\partial_R Y_R\|_{C^0} \le C(\varepsilon)\|H\|\sigma^{-1/2}\sqrt{E}.$$

Consequently,

$$|F_H(R_0)| \le |I|^{-1}\int_I |F_H(R)|\, dR + |I| \sup_I |F_H'| \le C(\varepsilon)\|H\|\sigma^{1/2}\sqrt{E} \le C\|H\|\sqrt{E}.$$

The last inequality holds after decreasing ρ_0 in terms of ε, since $\sigma \le C\rho$. On a graphical annulus, (3.11) and (3.6) also show that moving either circle changes its flux by at most $C\|H\|\sqrt{E}$. □

Claim 2. On a neck with $\sigma > C(\varepsilon)\rho_*$, the parameter $a \in \mathbb{R}$ can be normalized so that

$$\int_{c_+} \eta_{P-I/3} = 4\pi a, \qquad \int_{c_-} \eta_{P-I/3} = -4\pi a + O(\sqrt{E}), \tag{3.12}$$

while preserving the conclusions of Lemma 3.7. For $\sigma \le C(\varepsilon)\rho_*$, the first identity defines a with $|a| \le C(\varepsilon)\rho_*^2$. If $|q_+ - q_-| > C(\varepsilon)\sigma$, then $|a| \le C\sqrt{E}$.

Proof. For $\sigma > C(\varepsilon)\rho_*$, write $\tilde{a}$ for the real parameter of the Lawlor neck model in Lemma 3.7, and define a by the first identity in (3.12). Set $H = P - I/3$ and $h = \langle HF_s, F_s\rangle$. By (3.5) and (3.6), the linear part of the flux is $3\int (h\partial_\nu u_{R_0} - u_{R_0}\partial_\nu h)\, ds + O(r^{-1/2}\sqrt{E})$. Since $h(p_+) = 2/3$ and $dh(p_+) = 0$, the leading logarithmic term $\tilde{a}\log r$ in u_{R_0} contributes $3 \cdot (2/3) \cdot 2\pi\tilde{a} = 4\pi\tilde{a}$.

By Proposition 3.6, the contribution from the average zero part of u_{R_0} is bounded by $C(\varepsilon\sigma^2 + \rho^2 r\log(1/\rho) + r^{-1/2}\sqrt{E})$, and the quadratic nonlinear error is bounded by $C(\sigma^4/r^2 + \rho^4\log^2(1/\rho) + E/r^3)$. With $C(\varepsilon)$ in $r = C(\varepsilon)\sigma$ and $\sigma > C(\varepsilon)\rho_*$ chosen sufficiently large, the quadratic error is at most $\varepsilon\sigma^2$ for sufficiently small ρ. Hence

$$|a - \tilde{a}| \le C\varepsilon\sigma^2 + C(\varepsilon)\rho^2\sigma\log(1/\rho) + C\sigma^{-1/2}\sqrt{E} \le C\varepsilon\sigma^2.$$

Replacing $\tilde{a}$ by a changes the Lawlor neck model by only $O(\varepsilon\sigma)$ in the C^2_σ norm. With a sufficiently accurate initial approximation, the conclusions of Lemma 3.7 therefore remain valid. In particular, (3.4) holds with $\pm a$ on the corresponding $\pm$ ends. Claim 1 gives the second identity in (3.12).

When $\rho_* \le \sigma \le C(\varepsilon)\rho_*$, the coefficient bound in Proposition 3.6 and $E \le \rho_*^5$ give $|a| \le C(\sigma^2 + \sigma^{-1/2}\sqrt{E}) \le C(\varepsilon)\rho_*^2$. If $|q_+ - q_-| > C(\varepsilon)\sigma$, the two cutoff regions lie in disjoint balls of radius $O(C(\varepsilon)\sigma)$ about $q_\pm$. For each nearby slice used in Claim 1, their boundaries meet Γ_R only along $c_-(R)$ and $c_+(R)$, respectively. Since Γ_R is closed, its restriction to either region has boundary the corresponding circle. The Stokes and radial estimates in Claim 1 therefore apply to each circle separately. Taking $H = P - I/3$ gives

$$4\pi|a| = \left| \int_{c_+} \eta_{P-I/3} \right| \le C(\varepsilon)\sigma^{1/2}\sqrt{E} \le C\sqrt{E}.$$

□

3.8 The upper bound for the density excess

Recall Γ_R and put $\zeta = (\iota_x \operatorname{Re}\Omega)|_{S^5}$. Since each Λ_s is a closed minimal immersion, its area is independent of s. Thus (2.2) gives

$$3m(R) = \int_{\Gamma_R} \zeta - 3\Theta_0 = \int_{\Gamma_R} \zeta - \int_{\Lambda_s} \zeta.$$

Proposition 3.8 (Excess estimate)**.** *There are $\delta, C > 0$ such that, if (3.1) holds and $\mathbf{C}_s$ is a best approximating cone with $s \in \operatorname{int}\mathcal{M}$, then*

$$m(1) \le C(D^2 + E).$$

Proof. We shall prove, for almost every $R \in (5/4, 7/4)$, that

$$\left| \int_{\Gamma_R} \zeta - \int_{\Lambda_s} \zeta \right| \le C \int_{\Gamma_R} (\operatorname{dist}(x, \mathbf{C}_s)^2 + |\lambda|^2)\, d\|\Gamma_R\|. \tag{3.13}$$

By coarea,

$$\int_{5/4}^{7/4} \int_{\Gamma_R} (\operatorname{dist}(x, \mathbf{C}_s)^2 + |\lambda|^2)\, d\|\Gamma_R\|\, dR \le C(D^2 + E).$$

Selecting a good radius $R \in (5/4, 7/4)$ and using (3.13) and monotonicity, we obtain $m(1) \le m(R) \le C(D^2 + E)$.

We may assume that Γ_R is smooth. Put $\zeta' = (\iota_x \operatorname{Im}\Omega)|_{S^5}$. The argument uses a homotopy H_t, $0 \le t \le 1$, with the following properties on Γ_R:

$$H_1 = \mathrm{id}, \qquad (H_0)_\# \Gamma_R = \Lambda_s, \qquad H_0^* \zeta' = 0,$$
$$|\dot{H}_t(x)| \le C \operatorname{dist}(x, \mathbf{C}_s), \qquad |dH_t| \le C.$$

The differential bounds hold almost everywhere, with a uniform constant. We shall obtain H_t by short geodesic interpolation from a uniformly Lipschitz map H_0 to the identity; first we prove the estimate from these properties.

Step 1: compare the integrals. For a differential form ϑ, write

$$\mathcal{K}_t \vartheta = \int_0^t H_\tau^* (\iota_{\dot{H}_\tau} \vartheta)\, d\tau.$$

Since $H_0^*\zeta' = 0$ on Γ_R, the homotopy formula gives

$$H_t^*\zeta' = d\mathcal{K}_t\zeta' + \mathcal{K}_t(d\zeta'), \qquad \partial_t(H_t^*\lambda) = d(\lambda(\dot{H}_t)) + H_t^*(\iota_{\dot{H}_t}d\lambda).$$

Applying Stokes to the currents defined by the Lipschitz homotopy, and using $d\zeta = -3\lambda\wedge\zeta'$, the two identities above, and $\mathcal{K}_0 = 0$, we obtain

$$\begin{aligned}
&\int_{\Gamma_R}\zeta - \int_{\Lambda_s}\zeta \\
&= -3\int_0^1\int_{\Gamma_R}\left\{\lambda(\dot{H}_t)H_t^*\zeta' - H_t^*\lambda\wedge H_t^*(\iota_{\dot{H}_t}\zeta')\right\}dt \\
&= 3\int_0^1\int_{\Gamma_R}\left\{\partial_t(H_t^*\lambda\wedge\mathcal{K}_t\zeta') - H_t^*(\iota_{\dot{H}_t}d\lambda)\wedge\mathcal{K}_t\zeta' - \lambda(\dot{H}_t)\mathcal{K}_t(d\zeta')\right\}dt \\
&= 3\int_{\Gamma_R}\lambda\wedge\mathcal{K}_1\zeta' - 3\int_0^1\int_{\Gamma_R}\left\{H_t^*(\iota_{\dot{H}_t}d\lambda)\wedge\mathcal{K}_t\zeta' + \lambda(\dot{H}_t)\mathcal{K}_t(d\zeta')\right\}dt.
\end{aligned}$$

The second equality uses integration by parts on Γ_R and the product rule in t. Each factor involving $\dot{H}_t$ or $\mathcal{K}_t$ is bounded by $C\operatorname{dist}(x,\mathbf{C}_s)$. The time integral is therefore bounded by $C\int_{\Gamma_R}\operatorname{dist}(x,\mathbf{C}_s)^2\,d\|\Gamma_R\|$, and Young's inequality bounds the term at $t=1$ by the right-hand side of (3.13).

Step 2: construct the homotopy. Away from the points q_-, q_+ corresponding to the double points of Λ, we use the usual projection onto the nearby sheet. Near q_- and q_+, we join the two projections continuously by allowing the image to move along the short Reeb segment between q_- and q_+. When the two points coincide, this segment shrinks to their intersection point. We shall construct a uniformly Lipschitz retraction Π_s from a fixed neighbourhood of Λ_s onto this union of sheets and segments. Since it fixes Λ_s, it satisfies

$$|\Pi_s(x) - x| \le C\operatorname{dist}(x,\Lambda_s).$$

Let $\beta \ge 0$ be the length of the short Reeb segment between q_- and q_+. Choose local coordinates $(x,y,\tau)\in\mathbb{R}^2\times\mathbb{R}^2\times\mathbb{R}$ in which the two local sheets of Λ_s are the local coordinate planes $\{(x,0,-\beta/2)\}$ and $\{(0,y,\beta/2)\}$, and the Reeb segment is $\{(0,0,\tau) : |\tau|\le\beta/2\}$. Set $w = \tau + |y| - |x|$ and define

$$\Pi_s(x,y,\tau) = \begin{cases}
\left(\dfrac{-2w-\beta}{\max\{\sqrt{|x|^2+|y|^2}, -2w-\beta\}}x, 0, -\beta/2\right), & w < -\beta/2, \\
(0,0,w), & |w|\le\beta/2, \\
\left(0, \dfrac{2w-\beta}{\max\{\sqrt{|x|^2+|y|^2}, 2w-\beta\}}y, \beta/2\right), & w > \beta/2.
\end{cases}$$

The three cases take values in the first plane, the joining segment, and the second plane, respectively. In particular,

$$\Pi_s(x,y,\tau) = \begin{cases}(x,0,-\beta/2), & |x| > 3(|y| + |\tau+\beta/2|), \\ (0,y,\beta/2), & |y| > 3(|x| + |\tau-\beta/2|).\end{cases}$$

This is a retraction onto the two planes and their joining segment, with Lipschitz constant independent of β.

Away from fixed neighbourhoods of these points, extend Π_s by a tubular projection onto Λ_s. On each overlap, both maps take values in the same embedded portion of Λ_s. Interpolate between them in local coordinates on this surface, using fixed cutoffs. This gives a retraction on a neighbourhood of fixed width around all of Λ_s, with Lipschitz constant uniform in s.

By (3.1), Γ_R is supported in this fixed neighbourhood. It is easy to see that

$$(\Pi_s)_{\#}\Gamma_R = \Lambda_s.$$

On the sheets ζ' vanishes, while on the joining segments Π_s has rank at most one almost everywhere. Consequently $\Pi_s^*\zeta' = 0$ almost everywhere on Γ_R.

The short geodesic homotopy from Π_s to the identity has the required endpoint properties and satisfies

$$|\dot{H}_t(x)| \le C\,\mathrm{dist}(x,\Lambda_s) \le C\,\mathrm{dist}(x,\mathbf{C}_s), \qquad |dH_t| \le C.$$

Thus it has all the properties used in Step 1. □

4 The gluing obstruction

Work on Γ_{R_0}, where R_0 is the regular radius from Proposition 3.5. Lemma 3.7 gives extracted necks whose largest stopping radius is comparable to ρ, with $\rho \ge C^{-1}D^{2/\gamma}$. We shall prove the discrete Łojasiewicz inequality in Proposition 4.6 by combining the loop periods, flux balance and variation of the Reeb chord actions with the analytic gradient inequality.

Orient the edges of $G(\Lambda)$ and write $a_e, \sigma_e, q_{e,\pm}(s), c_{e,\pm}, P_e(s)$ for the corresponding local quantities. Let $G' = (V, E')$ be the subgraph of $G(\Lambda)$ containing all vertices and exactly the edges e with $\sigma_e > C(\varepsilon)\rho_*$, for a sufficiently large $C(\varepsilon)$, to be chosen in the proof. The short Reeb chord from $q_{e,-}(s)$ to $q_{e,+}(s)$ has signed action $\beta_e(s)$, so $q_{e,+}(s) = e^{i\beta_e(s)}q_{e,-}(s)$. Here $\beta_e(0) = 0$, but $\beta_e(s)$ need not vanish. The coefficient comparison and Lemma 3.7 give

$$\begin{aligned} &|a_e| \asymp \sigma_e^2, \quad |\beta_e(s)| \le C\sigma_e && (e \in E'),\\ &|a_e| \le C(\varepsilon)\rho_*^2 && (e \notin E'). \end{aligned}$$

For $e \notin E'$, the stopping construction gives $\sigma_e \le C(\varepsilon)\rho_*$. The separation estimate in Claim 2 after Lemma 3.7 and $|q_{e,+} - q_{e,-}| = 2|\sin(\beta_e(s)/2)|$ for the short Reeb chord therefore give

$$|a_e| > C\sqrt{E} \quad\Longrightarrow\quad |q_{e,+} - q_{e,-}| \le C(\varepsilon)\sigma_e \le C(\varepsilon)\rho_* \quad\Longrightarrow\quad |\beta_e(s)| \le C(\varepsilon)\rho_*. \tag{4.1}$$

Proposition 3.5 also gives $\max_{e\in E'}|a_e| \asymp \rho^2$.

For an oriented simple cycle γ_j in $G(\Lambda)$, let $\xi_e(\gamma_j)$ be 1, -1, or 0 according as γ_j traverses e with its orientation, against it, or not at all.

4.1 The logarithmic term in loop periods

On Γ_{R_0}, for each extracted edge e, orient the neck and its Reeb chord c_s from the minus branch to the plus branch, so $\beta_e(s) = \int_{c_s}\lambda$. An oriented simple cycle γ_j in G' is represented by a loop γ in Γ_{R_0} crossing its Lawlor necks and joining their ends in the graphical region, avoiding all other puncture neighbourhoods.

Proposition 4.1 (Loop periods)**.** *There is $C > 0$ such that every oriented simple cycle γ_j in G' has a representative γ as above satisfying*

$$\int_\gamma \lambda = \sum_{e\in E'} \xi_e(\gamma_j)\{\beta_e(s) + 2a_e\log|a_e|\} + e_\gamma, \tag{4.2}$$

where

$$\begin{aligned} |e_\gamma| &\le C\varepsilon\rho^2\log(1/\rho),\\ \left|\int_\gamma \lambda\right| &\le C(\varepsilon)\rho_*^{-1/2}\sqrt{E}. \end{aligned}$$

Proof. Step 1: choose the loop and bound its period. For each vertex of γ_j, choose a path in the corresponding component of Σ_{r_0} joining the two boundary circles associated with the adjacent edges. Extend each end along a curve of constant θ down to $r = C(\varepsilon)\sigma$, using the stopping radius of that neck. Map these paths to Γ_{R_0} by the graphical parametrization and join each pair of endpoints through the corresponding Lawlor neck. Write γ_{core} for this segment between the two circles $r = C(\varepsilon)\sigma$, and γ_{in} and γ_{out} for the adjacent portions in the graphical annuli $C(\varepsilon)\sigma < r < r_0$ (Figure 1). The resulting loop γ inherits its orientation from γ_j. The paths in Σ_{r_0} can be chosen with uniformly bounded length. By (3.6),

$$\int_{\gamma_{\mathrm{in}}} |\lambda| \le C\sqrt{E} \int_{C(\varepsilon)\sigma}^{r_0} r^{-3/2}\, dr \le C\sigma^{-1/2}\sqrt{E}.$$

The same bound holds on γ_{out}. On γ_{core}, interior estimates give the bound with constant $C(\varepsilon)$; this segment has length $O(C(\varepsilon)\sigma)$. On Σ_{r_0} the bound is $C\sqrt{E}$. There are finitely many edges and $\sigma > C(\varepsilon)\rho_*$, so summing these bounds proves the second estimate. Moreover, $E \le \rho_*^5$ gives $\rho_*^{-1/2}\sqrt{E} \le \rho_*^2 \le \rho^2$. Thus the total contribution of the graphical paths is $O(\rho^2)$.

Step 2: compute the integral along γ_{core}. Fix an extracted Lawlor neck corresponding to $e \in E'$, with stopping radius σ, and work near its ends in the coordinates (p, ξ, z) of Subsection 3.6. Let x_-, x_+ be the endpoints of γ_{core} on the two circles $r = C(\varepsilon)\sigma$, with z-coordinates $u_\pm$. Let $\bar{x}_\pm \in \Lambda_s$ be their projections to the zero section. Join $\bar{x}_\pm$ to $x_\pm$ keeping p fixed. Since $dp = 0$, the λ-integral along each segment is $2u_\pm$. The path from $\bar{x}_-$ to $\bar{x}_+$ along the Legendrian branches and the Reeb chord has integral $\beta_e(s)$. Together with γ_{core}, these paths form a closed loop of length $O(C(\varepsilon)\sigma)$. Coning this loop to x_- gives a spanning surface of area $O(C(\varepsilon)\sigma^2)$. The Stokes formula and $d\lambda = 2\omega$ therefore give

$$\int_{\gamma_{\mathrm{core}}} \lambda = 2u_+ - 2u_- + \beta_e(s) + O(C(\varepsilon)\sigma^2).$$

By the comparison following Lemma 3.7, (3.4) holds with $\pm a_e$ on the corresponding $\pm$ ends. Since $|a_e| \asymp \sigma^2 \le C\rho^2$,

$$4a_e \log(C(\varepsilon)\sigma/r_0) = 2a_e \log|a_e| + O(C(\varepsilon)\sigma^2) = 2a_e \log|a_e| + O(C(\varepsilon)\rho^2).$$

Integrating (3.4) from r_0 to $C(\varepsilon)\sigma$ and using the bounds for u_{R_0} and for $u_{R_0} - \bar{u}_{R_0}$ in Proposition 3.6, we obtain

$$\begin{aligned} |2u_+ - 2u_- - 2a_e \log|a_e|| &\le C\varepsilon|a_e| \log(1/|a_e|) + C(\varepsilon)\rho^2 \\ &\le C\varepsilon\rho^2 \log(1/\rho). \end{aligned}$$

In the last inequality, we take ρ sufficiently small depending on ε to absorb $C(\varepsilon)\rho^2$ into $\varepsilon\rho^2 \log(1/\rho)$. Summing the resulting core integrals with their traversal signs and adding the $O(\rho^2)$ graphical contribution from Step 1 gives (4.2) and the stated error bound. □

4.2 Reeb chord actions and flux balance

For each edge e, let $p_{e,\pm} \in \Sigma$ be the corresponding preimages of the points $q_{e,\pm}(s) = F_s(p_{e,\pm}) \in \Lambda_s$.

Proposition 4.2 (Action variation and flux balance)**.** *A tangent vector $\dot{s} \in T_s\mathcal{M}$ represented by the Jacobi normal field $2hJF_s + J\nabla h$ satisfies*

$$d\beta_e(s)[\dot{s}] = 2\{h(p_{e,+}) - h(p_{e,-})\}. \tag{4.3}$$

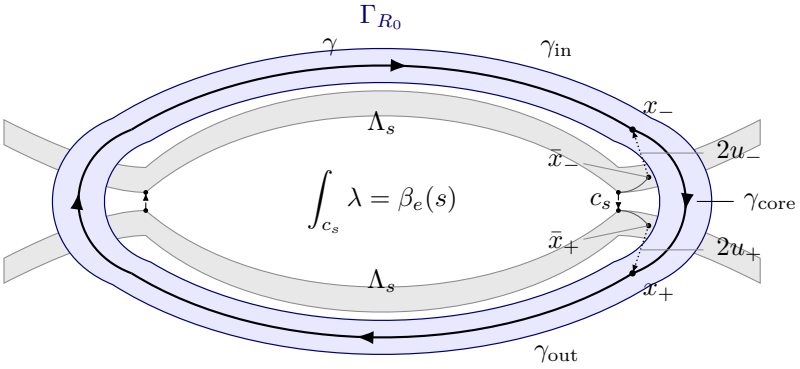


Figure 1: The loop γ on Γ_{R_0} (blue) and the comparison surface Λ_s (grey). The endpoints $x_\pm$ lie at $r = C(\varepsilon)\sigma$. The dotted paths have λ-integrals $2u_\pm$; the short dashed Reeb chords are not part of γ.

The coefficients a_e satisfy

$$\left\| \sum_{e \in E} a_e \, d\beta_e(s) \right\|_{T_s^*\mathcal{M}} \leq C(\rho^4 \log(1/\rho) + \sqrt{E}). \tag{4.4}$$

At every vertex v,

$$\left\| \sum_{e \text{ enters } v} a_e(P_e(s) - I/3) - \sum_{e \text{ leaves } v} a_e(P_e(s) - I/3) \right\| \leq C(\rho^4 \log(1/\rho) + \sqrt{E}). \tag{4.5}$$

Proof. Step 1: variation of the action. Since the restriction of $d\lambda$ to S^5 annihilates the Reeb direction, variation of the chord integral leaves only the endpoint terms. Their tangential parts vanish on the Legendrian branches, giving (4.3).

Step 2: flux balancing. Take a Jacobi function h with $\|h\|_{C^3(\Sigma)} \leq 1$, and let σ_l be the cutoff radius chosen at the puncture $p_l \in \Sigma$. Rearranging the slice equation from Step 1 of Proposition 3.6 and absorbing the nonlinear terms into $\mathcal{N}$ gives, on $\Sigma \setminus \bigcup_l B^\Sigma_{C\sigma_l}(p_l)$ with C sufficiently large and independent of ε,

$$\Delta_\Sigma u_{R_0} + 6u_{R_0} = \mathcal{N}, \qquad |\mathcal{N}| \leq C\left(\frac{\sigma^4}{r^4} + \frac{\rho^4}{r^2} + r^{-5/2}\sqrt{E}\right).$$

The graphical estimate $|u_{R_0}| + |du_{R_0}| + |D^2 u_{R_0}| \leq C(\sigma^2/r^2 + \rho^2/r)$ ensures smallness for $r \geq C\sigma$, since $\sigma \geq \rho^p$ and $p < 2$. Squaring the right-hand side bounds the quadratic Taylor remainder and gives the first two terms in the bound for $\mathcal{N}$. The radial and Liouville-form terms give the last term, as in Step 1 of Proposition 3.6. In particular, $|\mathcal{N}| \leq C(\rho^4 + \sqrt{E})$ on Σ_{r_0}.

Let H_l be trace-free Hermitian matrices with $\|H_l\| \leq C$ such that $h_l = \langle H_l F_s, F_s \rangle$ agrees with h to first order at p_l. Then $|h - h_l| + r|d(h - h_l)| \leq Cr^2$. Apply Green's formula to the

Jacobi function $h - h_l$ on $B^{\Sigma}_{r_0}(p_l) \setminus B^{\Sigma}_{C\sigma_l}(p_l)$. The graphical bounds give

$$\begin{aligned}
&\left| \int_{r=r_0} \left((h-h_l)\partial_r u_{R_0} - u_{R_0}\partial_r(h-h_l) \right) ds \right| \\
&\quad \le \left| \int_{B^{\Sigma}_{r_0}(p_l)\setminus B^{\Sigma}_{C\sigma_l}(p_l)} (h-h_l)\mathcal{N}\, d\mathrm{Area} \right| \\
&\qquad + \left| \int_{r=C\sigma_l} \left((h-h_l)\partial_r u_{R_0} - u_{R_0}\partial_r(h-h_l) \right) ds \right| \\
&\quad \le C \int_{C\sigma_l}^{r_0} \left(\frac{\sigma_l^4}{r} + \rho^4 r + r^{1/2}\sqrt{E} \right) dr + C(\sigma_l^4 + \rho^2\sigma_l^2 \log(1/\sigma_l)) \\
&\quad \le C(\rho^4 \log(1/\rho) + \sqrt{E}),
\end{aligned} \tag{4.6}$$

where the last inequality uses $\rho_* \le \sigma_l \le C\rho$. Now apply Green's formula to h on Σ_{r_0}:

$$\sum_l \int_{r=r_0} (h\partial_r u_{R_0} - u_{R_0}\partial_r h)\, ds = -\int_{\Sigma_{r_0}} h\mathcal{N}\, d\mathrm{Area} = O(\rho^4 + \sqrt{E}).$$

By (4.6), replacing h by h_l in each boundary integral changes the sum by at most $C(\rho^4 \log(1/\rho) + \sqrt{E})$. Pull back the flux forms by the graphical parametrization and orient each circle as the boundary of $B^{\Sigma}_{r_0}(p_l)$. Then the $O(\rho^2)$ graphical bound at $r = r_0$, together with (3.5) and (3.6), gives

$$\int_{r=r_0} (h_l\partial_r u_{R_0} - u_{R_0}\partial_r h_l)\, ds = \frac{1}{3}\int_{r=r_0} \eta_{H_l} + O(\rho^4 + \sqrt{E}).$$

Consequently,

$$\left| \sum_l \int_{\partial B^{\Sigma}_{r_0}(p_l)} \eta_{H_l} \right| \le C(\rho^4 \log(1/\rho) + \sqrt{E}). \tag{4.7}$$

Step 3: pair the endpoint fluxes. We evaluate the flux integrals in (4.7) in pairs, choosing the two matrices from Step 2 at each edge simultaneously. Fix an edge e, with $p_\pm \in \Sigma$ and $q_\pm = F_s(p_\pm)$, and recall $P = q_- q_-^* = q_+ q_+^*$. Write $h_\pm = \langle H_\pm F_s, F_s\rangle$ for the two functions h_l. The requirements on h_+ and h_- are

$$h_\pm(p_\pm) = h(p_\pm), \qquad dh_-(p_\pm) = dh_+(p_\pm) = dh(p_\pm).$$

Here the differentials are taken on Λ_s and do not include the radial direction of the Lagrangian cone. The transversality of the two tangent planes in $(\mathbb{C}q_-)^\perp$ allows us to choose H_- so that $dh_-(p_\pm) = dh(p_\pm)$, while independently prescribing $h_-(p_-) = h(p_-)$. Since $q_\pm$ lie on the same Hopf fibre, $h_-(p_+) = h(p_-)$. The function $\frac{3}{2}\langle (P - I/3)F_s, F_s\rangle$ has value 1 and zero differential at $p_\pm$. Hence the choice

$$H_+ = H_- + \frac{3}{2}\{h(p_+) - h(p_-)\}(P - I/3)$$

satisfies the stated requirements. The uniform transversality of the two tangent planes gives $\|H_-\| + \|H_+\| \le C$, with C independent of s.

By Claim 1 in the flux interpretation following Lemma 3.7, moving the circles from $r = C(\varepsilon)\sigma_e$ to $r = r_0$ changes each flux by $O(\sqrt{E})$. Thus (3.10) and (3.12) give the contribution of this edge to (4.7) as

$$6\pi a_e\{h(p_+) - h(p_-)\} + O(\sqrt{E}).$$

Restoring the edge indices, summing and rescaling h gives

$$\left|\sum_{e\in E} a_e\{h(p_{e,+}) - h(p_{e,-})\}\right| \le C\|h\|_{C^3(\Sigma)}(\rho^4 \log(1/\rho) + \sqrt{E}). \tag{4.8}$$

Fix a smooth norm on $T\mathcal{M}$. For $\|\dot{s}\| = 1$, the corresponding Jacobi functions have uniformly bounded $C^3(\Sigma)$ norm by elliptic estimates. Combining (4.3) with (4.8) gives (4.4).

Step 4: balance at each vertex. Fix a vertex v and a trace-free Hermitian matrix H. Consider the Jacobi function defined by $h = \langle HF_s, F_s\rangle$ on Σ_v and $h = 0$ on the other components. It satisfies $\|h\|_{C^3(\Sigma)} \le C\|H\|$. The points $q_{e,-}(s)$ and $q_{e,+}(s)$ have the same projection matrix $P_e(s)$, so

$$\sum_{e\in E} a_e\{h(p_{e,+}) - h(p_{e,-})\} = \sum_{e \text{ enters } v} a_e \operatorname{tr}(HP_e(s)) - \sum_{e \text{ leaves } v} a_e \operatorname{tr}(HP_e(s)).$$

An edge from v to itself contributes to both sums with opposite signs. The flux balance (4.8) bounds this expression by $C\|H\|(\rho^4 \log(1/\rho) + \sqrt{E})$ in absolute value. Since $\operatorname{tr} H = 0$, we may replace $P_e(s)$ by $P_e(s) - I/3$. Taking the supremum over $\|H\| \le 1$ controls the norm of the trace-free matrix in (4.5), proving the bound. □

4.3 Application of the Łojasiewicz gradient inequality

Lemma 4.3 (Weighted Łojasiewicz inequality)**.** *Let γ_j be distinct oriented simple cycles of $G(\Lambda)$, and put $g_j(s) = \sum_{e\in\gamma_j} \xi_e(\gamma_j)\beta_e(s)$. There are $C > 0$ and $0 < \theta < 1$ such that, for sufficiently small s and real coefficients $|w_j| \le 1$,*

$$\left|\sum_j w_j g_j(s)\right|^\theta \le C\left(\left\|\sum_j w_j\, dg_j(s)\right\| + \left(\sum_j |w_j|\,|g_j(s)|^2\right)^{1/2}\right).$$

The constants are uniform over the choices of cycles and orientations.

Proof. By Jacobi integrability, we may take Λ_s to depend real analytically on s. Since the two local sheets intersect transversely after Hopf projection, the short Reeb chords and their actions depend analytically on s. Thus each g_j is analytic and $g_j(0) = 0$.

For fixed signs $\epsilon_j \in \{-1, 1\}$, set $f(s, z) = \sum_j \epsilon_j z_j^2 g_j(s)$, with s, z independent. Since $f(0, z) = 0$, the Łojasiewicz gradient inequality and compactness give

$$|f(s, z)|^\theta \le C(\|d_s f(s, z)\| + |d_z f(s, z)|)$$

uniformly for small s and $|z_j| \le 1$. Writing $w_j = \epsilon_j z_j^2$ gives the stated inequality. There are only finitely many choices of cycles, orientations and signs, so C and θ can be chosen uniformly. □

Proposition 4.4 (Gluing obstruction)**.** *Assume that Λ satisfies the intersection condition in Definition 1.1, and let θ be as in Lemma 4.3. Suppose*

$$\max\{20/3, 4 + 4\theta\} < q_0 < 8.$$

Then there are $\delta, \rho_0 > 0$ such that, if (3.1) holds and the stopping radius ρ supplied by Proposition 3.5 satisfies $0 < \rho < \rho_0$, then

$$E \ge \rho^{q_0}. \tag{4.9}$$

Proof. We will choose $1 < p < q_0/5$ at the end of the proof. If $E \geq \rho^{q_0}$, there is nothing to prove. We therefore assume $E < \rho^{q_0} < \rho^{5p} = \rho_*^5$, so the preceding neck construction applies. Write $\Delta = \rho^4 \log(1/\rho) + \sqrt{E}$; then $\Delta = o(\rho^{2p})$ as $\rho \to 0$.

Step 1: reduce the vertex balance to cycle coefficients. Choose a spanning forest of $G(\Lambda)$. Adding each omitted edge produces a simple cycle; no two of these cycles share an edge by the intersection condition in Definition 1.1. Denote them by γ_j and orient their edges cyclically, so $\xi_e(\gamma_j) = 1$ for $e \in \gamma_j$.

Claim. Let e_j be the edge of γ_j outside the spanning forest. Then

$$|a_e - a_{e_j}| \leq C\Delta \quad (e \in \gamma_j), \qquad |a_e| \leq C\Delta \quad (e \text{ a bridge}).$$

To prove the first estimate, remove an edge $e \in \gamma_j \setminus \{e_j\}$ from the forest. This splits its tree into two components; let S be the vertex set of the component into which e points. Only e and e_j join S to its complement in $G(\Lambda)$, because any further edge, together with the unique path in the original forest between its endpoints, would produce a second simple cycle containing e, contrary to the intersection condition in Definition 1.1.

Sum the vertex balances (4.5) over S. Every edge with both endpoints in S contributes its matrix once with each sign, so all such terms cancel exactly. Since γ_j is oriented cyclically, e enters S and e_j leaves it. Consequently,

$$\begin{aligned} &a_e(P_e - I/3) - a_{e_j}(P_{e_j} - I/3) \\ &\quad = \sum_{v \in S} \left(\sum_{f \text{ enters } v} a_f(P_f - I/3) - \sum_{f \text{ leaves } v} a_f(P_f - I/3) \right) = O(\Delta). \end{aligned}$$

The number of vertices is fixed, so the sum of the errors is still $O(\Delta)$. Since the middle eigenvalue of $a(P - I/3)$ is $-a/3$ for either sign of a, eigenvalue continuity gives the first estimate. Removing a bridge from $G(\Lambda)$ separates its endpoints. Summing over the component containing one endpoint leaves only that edge in the boundary sum, giving the estimate.

Set $t_j = a_{e_j}$, so $a_e = t_j + O(\Delta)$ on γ_j and $|t_j| \leq C\rho^2$. Discard the bridges and cycles with $|t_j| \leq C\Delta$, for a sufficiently large fixed C. Their contribution to (4.4) is $O(\Delta)$. All sums below are over the remaining cycles, with g_j as in Lemma 4.3. Replacing a_e by t_j in (4.4) gives

$$\left\| \sum_j t_j \, dg_j(s) \right\| \leq C\Delta. \tag{4.10}$$

Each g_j includes all edges of its cycle.

Step 2: estimate the weighted sums of chord actions. On each remaining cycle, $|a_e| \asymp |t_j|$ for every $e \in \gamma_j$. We distinguish two cases.

If $\gamma_j \subset E'$, the loop period formula in Proposition 4.1 gives

$$g_j(s) = 2\#\gamma_j \, t_j \log(1/|t_j|) + O\Big(\varepsilon\rho^2 \log(1/\rho)\Big). \tag{4.11}$$

Here replacing a_e by t_j costs $O(\Delta \log(1/\rho))$, and the loop period is bounded by $C(\varepsilon)\rho_*^{-1/2}\sqrt{E}$. Since $E < \rho^{5p}$ and $p > 1$, both are $o(\rho^2 \log(1/\rho))$ and are absorbed by taking ρ sufficiently small in terms of ε.

If $\gamma_j \not\subset E'$, an edge outside E' has coefficient $O(C(\varepsilon)\rho_*^2)$, so $|t_j|$ and all $|a_e|$ on this cycle are $O(C(\varepsilon)\rho_*^2)$. Hence $|\beta_e(s)| \leq C(\varepsilon)\rho_*$ for every edge: for $e \in E'$ this follows from $|\beta_e(s)| \leq C\sqrt{|a_e|}$; for $e \notin E'$, it follows from (4.1), since $|a_e| > C\sqrt{E}$ on every remaining cycle. Thus

$$|t_j| \leq C(\varepsilon)\rho^{2p}, \qquad |g_j(s)| \leq C(\varepsilon)\rho^p \qquad (\gamma_j \not\subset E'). \tag{4.12}$$

Finally, since $\max_{e\in E'}|a_e| \asymp \rho^2$ and $\Delta = o(\rho^2)$, at least one remaining cycle has $|t_j| \asymp \rho^2$. By (4.12) and $p > 1$, this cycle is contained in E'.

The cycles not contained in E' contribute $O(C(\varepsilon)\rho^{3p-2})$ to $\rho^{-2}\sum_j t_j g_j(s)$ by (4.12). For the other cycles, (4.11) gives

$$\rho^{-2}\sum_j t_j g_j(s) = 2\rho^{-2}\sum_{\gamma_j\subset E'} \#\gamma_j\, t_j^2 \log(1/|t_j|) + O\Big(\varepsilon\rho^2\log(1/\rho) + C(\varepsilon)\rho^{3p-2}\Big).$$

For small ρ, each summand $\#\gamma_j\, t_j^2\log(1/|t_j|)$ on the right is positive, since $0 < |t_j| \le C\rho^2 < 1$. The cycle with $|t_j| \asymp \rho^2$ contributes at least $C^{-1}\rho^2\log(1/\rho)$ in absolute value. If $p > 4/3$, then $C(\varepsilon)\rho^{3p-2} = o(\rho^2)$ for each fixed ε. Choosing ε and then ρ small therefore gives

$$\left|\rho^{-2}\sum_j t_j g_j(s)\right| \ge C^{-1}\rho^2\log(1/\rho).$$

The same estimates also give

$$\rho^{-1}\left(\sum_j |t_j|\,|g_j(s)|^2\right)^{1/2} \le C\rho^2\log(1/\rho) + C(\varepsilon)\rho^{2p-1}.$$

Step 3: apply the weighted Łojasiewicz inequality. Apply Lemma 4.3 with $w_j = t_j/\max_k|t_k|$. Since $\max_k|t_k| \asymp \rho^2$, it gives

$$\begin{aligned}(\rho^2\log(1/\rho))^\theta &\le C\left(\rho^{-2}\Delta + \rho^2\log(1/\rho)\right) + C(\varepsilon)\rho^{2p-1}\\ &\le C\left(\rho^2\log(1/\rho) + \rho^{q_0/2-2}\right) + C(\varepsilon)\rho^{2p-1},\end{aligned}$$

by (4.10), Step 2 and $E < \rho^{q_0}$. We now choose

$$\max\{4/3, \theta+1/2\} < p < q_0/5.$$

The assumed range of q_0 and $\theta < 1$ make this interval nonempty. With p fixed, choose ε sufficiently small as in Step 2. Then $C(\varepsilon)$ is fixed, and $2p-1 > 2\theta$ and $q_0/2-2 > 2\theta$ make the right-hand side $o((\rho^2\log(1/\rho))^\theta)$, a contradiction for small ρ. All smallness requirements are met by choosing δ, ρ_0 sufficiently small. Hence $E \ge \rho^{q_0}$, proving (4.9). □

Example 4.5 (The two-plane case). The two Reeb chords are antipodal images. Since $x \mapsto -x$ preserves λ, their actions agree when oriented from the first sphere to the second. The cycle traverses them in opposite directions, so $g(s) = 0$ for every s. The flux and period estimates then give the obstruction directly.

4.4 The discrete Łojasiewicz estimate

Proposition 4.6 (Discrete Łojasiewicz estimate)**.** *Assume that Λ satisfies the intersection condition in Definition 1.1. There are $q \in (1,2)$ and $\delta, C > 0$ such that, whenever $L_{0,R} = (\lambda_{0,R})_\# L$ satisfies (3.1),*

$$D(R)^{2q} \le CE(R).$$

If $\mathbf{C}_0$ is a tangent cone of L at the origin, then also

$$m(R)^q \le C\{m(4R) - m(R/4)\}. \tag{4.13}$$

Proof. By rescaling, it suffices to take $R = 1$. Fix q_0 as in Proposition 4.4, and choose

$$q_0/2 < \gamma < 4, \qquad q = q_0/\gamma \in (1, 2).$$

We may assume $D > 0$ and $E \le \varepsilon_1 D^2$, with ε_1 as in Proposition 3.5; otherwise the estimate is immediate for $D \le 1$. Proposition 3.5 gives $\rho \ge C^{-1} D^{2/\gamma}$. Allard regularity at fixed scales ensures that ρ is below the threshold in Proposition 4.4 once δ is sufficiently small. That proposition gives

$$E \ge \rho^{q_0} \ge C^{-1} D^{2q}.$$

When $\mathbf{C}_0$ is a tangent cone, Proposition 3.8 and the uniform bound for E give

$$m(1)^q \le C(D^{2q} + E^q) \le CE = C\{m(4) - m(1/4)\}.$$

Rescaling proves (4.13). □

Remark. The strict inequality $q < 2$ is essential for the radial summability in Lemma 5.1. Since $q = q_0/\gamma$ and $\gamma < 4$, we need $q_0 < 2\gamma < 8$. If only $E \ge \rho^8$ were available, the same argument would give $m(1)^{8/\gamma} \le CE$, with exponent greater than 2. The bound $O(\rho^4 \log(1/\rho))$ on the flux error suffices for the argument in Step 3 of Proposition 4.4 with any Łojasiewicz exponent $\theta < 1$. A weaker bound $O(\rho^3)$ need not suffice when $\theta > 1/2$.

Proof of Theorem 1.4. Let Λ' be a special Legendrian integral two-cycle sufficiently close to Λ in the varifold topology. Varifold convergence and monotonicity give (3.1) for $\mathbf{C}(\Lambda')$. Since $E = 0$, Proposition 4.6 gives $D = 0$, so the cone is supported on $\mathbf{C}_s$. The initial graphical description implies multiplicity one on a regular patch of each component, and constancy gives $\mathbf{C}(\Lambda') = \mathbf{C}_s$ as currents. Taking radial slices gives $\Lambda' = \Lambda_s$. □

5 Uniqueness of the tangent cone

We use Proposition 4.6 to prove summability of the radial variation, then show that all sufficiently small rescalings satisfy (3.1). Compare [4, Section 1] and [16, Theorem 6.7].

5.1 Summability of radial variation

Let $R_j = 4^{-j} R_0$ and $\alpha = 1 - q/2 > 0$. All statements initially concern a finite range of scales on which (4.13) is known; no continuation below the last scale is assumed.

Lemma 5.1. *If $m(4R_0) \le 1$ and (4.13) holds at R_j for $0 \le j \le n$, then*

$$\sum_{j=0}^{n} \sqrt{m(4R_j) - m(R_j/4)} \le C m(4R_0)^{\alpha}. \tag{5.1}$$

The estimate also holds when the sum starts at any later scale R_j.

Proof. Write $u = m(4R)$ and $v = m(R/4)$. If $v \le u/2$, then $\sqrt{u - v} \le \sqrt{u} \le u^{\alpha} \le C(u^{\alpha} - v^{\alpha})$. Otherwise $m(R) \ge v > u/2$, so (4.13) gives $u - v \ge C^{-1} u^q$. Concavity then gives $u^{\alpha} - v^{\alpha} \ge \alpha u^{-q/2}(u - v) \ge C^{-1}\sqrt{u - v}$. Taking a telescoping sum gives the result. □

5.2 Proof of tangent-cone uniqueness

Choose $\delta > 0$ small enough that Proposition 4.6 applies and (3.1) ensures that a best approximating cone can be chosen with $s \in \operatorname{int} \mathcal{M}$. This follows from the minimizing property and the normal graphs on fixed regular regions, as at the beginning of Section 3.

For the argument below, measure height from the fixed cone by

$$H(R)^2 = R^{-5} \int_{L \cap (B_{8R} \setminus \overline{B}_{R/8})} \operatorname{dist}(x, \mathbf{C}_0)^2 \, d\|L\|(x).$$

By a stationarity calculation similar to that in Step 1 of Proposition 3.5, we obtain the Poincaré estimate

$$|H(t) - H(R)| \le C\sqrt{m(8R) - m(t/8)}, \qquad R/4 \le t \le R.$$

Choose $\tau > 0$ so that $C\tau^4 < \delta^2/2$. On the regular region where $\operatorname{dist}(x, \mathcal{R}_0) > \tau|x|$, choose $0 < \varepsilon < 1$ sufficiently small that the graphicality criterion at the beginning of Section 3 gives graphs of C^1 norm less than τ, and require

$$C\varepsilon^{2\alpha} < \delta^2/2, \qquad 64\varepsilon < \delta.$$

Finally, choose R_0 along the prescribed tangent sequence so that

$$H(R_0) < \varepsilon, \qquad m(16R_0) < \varepsilon,$$

and L_{0,R_0} satisfies (3.1) and is given by multiplicity-one normal graphs of C^1 norm less than ε over the regular sheets with $1/8 < |x| < 8$ and $\operatorname{dist}(x, \mathcal{R}_0) > (\tau/2)|x|$.

Inductive claim. The conditions (3.1) hold for every $R \in [R_n, R_0]$.

The starting conditions give the case $n = 0$. Suppose the claim holds down to R_n. For any $R \in [R_{n+1}, R_n]$, the Poincaré estimate and Lemma 5.1 give

$$H(R) \le H(R_0) + C \sum_{k=0}^{n} \sqrt{E(R_k)} + C\sqrt{\varepsilon} \le C\varepsilon^{\alpha}.$$

The finitely many endpoint terms arising from the larger annuli in H are bounded by $C\sqrt{\varepsilon}$, since $m(16R_0) < \varepsilon$. The same bound holds at every intermediate scale between R and R_0.

Using monotonicity and the inductive hypothesis, we apply Allard regularity to obtain normal graphs V over the regular region with $1/4 < |x| < 4$ and $\operatorname{dist}(x, \mathcal{R}_0) > \tau|x|$, satisfying

$$|V| + |\nabla V| \le C_\tau H(R) \le C_\tau \varepsilon^{\alpha} < \tau.$$

These graphs bound the reverse height on this region by $CH(R)^2$. The omitted part of $\mathbf{C}_0$ has mass $O(\tau^2)$ in $B_4 \setminus \overline{B}_{1/4}$, and each of its points lies within $C\tau$ of the adjacent graphs. Consequently,

$$\begin{aligned} &\int_{L_{0,R} \cap (B_4 \setminus \overline{B}_{1/4})} \operatorname{dist}(x, \mathbf{C}_0)^2 \, d\|L_{0,R}\|(x) \\ &\qquad + \int_{\mathbf{C}_0 \cap (B_4 \setminus \overline{B}_{1/4})} \operatorname{dist}(x, \operatorname{spt} L_{0,R})^2 \, d\|\mathbf{C}_0\|(x) \le CH(R)^2 + C\tau^4 \le C\varepsilon^{2\alpha} + C\tau^4. \end{aligned}$$

By the choices of τ and ε, this is less than δ^2. The mass condition follows from

$$\|L_{0,R}\|(B_4) - \|\mathbf{C}_0\|(B_4) = 64m(4R) \le 64\varepsilon < \delta.$$

This proves the inductive step.

Thus (3.1) holds at every sufficiently small scale. Proposition 4.6 places every tangent cone in the family $\mathbf{C}_s$, as in the proof of Theorem 1.4. Since δ may be chosen arbitrarily small, every tangent cone equals $\mathbf{C}_0$.

References

[1] W. K. Allard, *On the first variation of a varifold*, Ann. of Math. (2) **95** (1972), 417–491.

[2] W. K. Allard and F. J. Almgren, Jr., *On the radial behavior of minimal surfaces and the uniqueness of their tangent cones*, Ann. of Math. (2) **113** (1981), 215–265.

[3] T. C. Collins and Y. Li, *Uniqueness of some cylindrical tangent cones to special Lagrangians*, Geom. Funct. Anal. **33** (2023), 376–420.

[4] T. H. Colding and W. P. Minicozzi II, *On uniqueness of tangent cones for Einstein manifolds*, Invent. Math. **196** (2014), 515–588.

[5] N. Edelen and G. Székelyhidi, *A Liouville-type theorem for cylindrical cones*, Comm. Pure Appl. Math. **77** (2024), 3557–3580.

[6] M. Haskins and N. Kapouleas, *Special Lagrangian cones with higher genus links*, Invent. Math. **167** (2007), 223–294.

[7] R. Harvey and H. B. Lawson, Jr., *Calibrated geometries*, Acta Math. **148** (1982), 47–157.

[8] D. Joyce, *Lectures on special Lagrangian geometry*, in *Global Theory of Minimal Surfaces*, D. Hoffman (ed.), Clay Math. Proc. **2**, Amer. Math. Soc., 2005, 667–695.

[9] G. Lawlor, *The angle criterion*, Invent. Math. **95** (1989), 437–446.

[10] J. D. Lotay, F. Schulze and G. Székelyhidi, *Neck pinches along the Lagrangian mean curvature flow of surfaces*, arXiv:2208.11054 (2022).

[11] Y. Li and G. Székelyhidi, *Singularity formations in Lagrangian mean curvature flow*, Geom. Topol., to appear.

[12] P. Minter, *Regularity techniques in geometric measure theory*, Lecture notes, Princeton University, 2024.

[13] L. Simon, *Asymptotics for a class of non-linear evolution equations, with applications to geometric problems*, Ann. of Math. (2) **118** (1983), 525–571.

[14] L. Simon, *Cylindrical tangent cones and the singular set of minimal submanifolds*, J. Differential Geom. **38** (1993), 585–652.

[15] L. Simon, *Uniqueness of some cylindrical tangent cones*, Comm. Anal. Geom. **2** (1994), 1–33.

[16] G. Székelyhidi, *Uniqueness of certain cylindrical tangent cones*, Invent. Math. (2026), published online, doi:10.1007/s00222-026-01445-1.

[17] G. Székelyhidi, *Minimal hypersurfaces with cylindrical tangent cones*, J. Differential Geom. **133** (2026), 61–154.